

# Who Burns the Tokens? Fiscal Sustainability and Demand Concentration in GeoDePIN Networks

Zach Zukowski

Tokenization Systems

March 2026

## Abstract

This paper analyzes a class of DePIN protocols the paper terms GeoDePIN: networks whose value depends on where hardware is deployed, not merely that it exists. Location-dependence creates design pressures absent from compute-only DePIN alternatives, and the paper evaluates how protocols respond using on-chain evidence from 13 networks and a 34-month longitudinal case study of Helium. Helium became the first GeoDePIN protocol where paid usage fully offsets token emissions, with its Subsidy-to-Revenue ratio (S2R) rising from 0.013 under coverage-based rewards to 1.84 between May 2023 and February 2026. Data Credit burns now exceed HNT emissions by approximately 84%, producing net annual deflation of 3–4%. However, roughly five purchasers account for 90% of those burns: a demand concentration that aggregate fiscal metrics cannot detect. DIMO (S2R 4.24) subsequently crossed fiscal parity; GEODNET (0.62) and Hivemapper (0.183) occupy a growth-phase middle tier; the remaining protocols cluster near zero. Per a companion cross-sectional study, governance concentration in GeoDePIN tokens (Herfindahl-Hirschman Index 0.034 to 0.593) materially exceeds DeFi benchmarks (0.028 to 0.174). Four outputs organize the findings: a classification distinguishing GeoDePIN from location-independent protocols with testable predictions, a design playbook mapping mechanisms to protocol maturity stages, a health dashboard with demand-concentration guardrails, and a subtraction test for mechanism necessity.

**Keywords:** DePIN; GeoDePIN; Subsidy-to-Revenue ratio; burn-and-mint equilibrium; demand concentration; Helium; service verification; token velocity

**JEL Codes:** D43, D47, L14, L22, O33, R12

## List of Abbreviations

**VWAP** Volume-Weighted Average Price

**TWAP** Time-Weighted Average Price

**S2R** Subsidy-to-Revenue ratio

**RTK** Real-Time Kinematic (satellite positioning)

**HIP** Helium Improvement Proposal

**HHI** Herfindahl-Hirschman Index

**GeoDePIN** Geographically Distributed DePIN

**EIP** Ethereum Improvement Proposal

**DeVIN** Decentralized Virtual Infrastructure Network

**DePIN** Decentralized Physical Infrastructure Network

**DC** Data Credits (Helium network)

## 1. INTRODUCTION

### 1.1 The Coordination Problem

In mid-2024, Helium was generating roughly \$6,500 per month in paid usage across 975,000 IoT hotspots [60], [61]; revenue so thin that 99% of operator income came from token subsidies rather than customers. By Q3 2025, after a governance-led reform (HIP-147) shifted rewards from coverage to verified service, annualized paid usage had reached \$18.3M [28] and token burns exceeded emissions. The same network, the same hardware, a different rule: this is the institutional design problem DePIN poses. What mechanisms distinguish protocols that cross fiscal parity from those that remain subsidy-dependent? DePIN systems coordinate physical infrastructure through token incentives [22]; when the design is wrong, the failure modes are coverage mining, subsidy dependence, and governance capture.

GeoDePIN protocols constitute the most demanding stress test for institutional design [13] in the token economy: verification must prove physical presence under adversarial conditions, hardware lock-in raises the stakes of governance failure, and subsidy taper creates fiscal cliffs that force demand-driven sustainability.

### 1.2 GeoDePIN vs. DeVIN: A Taxonomic Distinction

A shared coordination pattern distinguishes these protocols from other DePIN projects: their networks derive competitive advantage from geographic coverage density across unique physical locations. We term this subcategory GeoDePIN (geographically distributed decentralized physical infrastructure) to distinguish it from DeVIN (decentralized virtual infrastructure networks) such as Filecoin [15]

(storage), Livepeer (video transcoding), or Render (GPU compute), where defensibility comes from compute density, throughput, or cost efficiency rather than location.

GeoDePIN projects offload capital expenditure to retail participants who bet that the network's future utility will justify their hardware investment. Four design pressures are distinctive to GeoDePIN: (1) Location is a first-class economic variable: the value of a Helium hotspot in downtown Manhattan differs from one in rural Wyoming by orders of magnitude. (2) Verification must prove physical presence, location authenticity, and service delivery under adversarial conditions, which is why Helium's HIP-147 transition to proof-of-service was decisive. (3) Subsidy taper threatens network contraction: when token rewards decline, operators in low-demand areas face negative unit economics first. (4) Incentives must be spatially intelligent: uniform global reward rates systematically over-reward saturated areas and under-reward frontier coverage.

The capability framework [48] interprets these design pressures as capability constraints: when operators absorb irrecoverable hardware sunk costs, their substantive freedom to exit the network is structurally negated, making governance voice mechanisms, not market exit, the primary channel for expressing dissatisfaction. This asymmetry is absent from DePIN protocols, where compute resources can be redirected across networks without capital loss, and it motivates the bicameral governance design developed in Section 2.

Network coverage models predict that GeoDePIN protocols face location-dependent tipping points absent from digital resource networks: Soria Ruiz-Ogarrio et al. [39] formalize optimal coverage equilibria for DePIN deployments, while Corn et al. [40] demonstrate empirically that spatial density determines forecasting accuracy in solar DePIN networks, confirming that geographic saturation imposes diminishing returns specific to location-dependent protocols.

The GeoDePIN design space can be summarized as the intersection of three optimization problems: location economics (where to deploy), verification integrity (how to prove real service), and subsidy discipline (when to taper). Section 6 develops these into a playbook; the taxonomy is formalized in Section 4.

### 1.3 Research Questions

Two research questions guide the analysis:

RQ1: What institutional mechanisms distinguish sustainable GeoDePIN protocols from fragile ones? Specifically, how do sink architecture, reward structure, and

governance design interact to determine whether a protocol achieves fiscal self-sufficiency?

RQ2: How should design playbooks differ for GeoDePIN (geographically distributed, retail-operated) versus DeVIN (compute-dense, institutionally operated) architectures? What testable predictions does this taxonomic distinction generate?

## 1.4 Paper Outline

Sections II through VII move from design architecture and methods through taxonomy, empirical evidence, and design playbook to conclusion. Practitioner and policy briefs appear as Supplementary Materials S1 and S2.

## 1.5 Positioning

DePIN has attracted growing attention from both industry and academic communities. Messari's annual DePIN reports [2], [28] track sector-level metrics (market capitalization, node counts, revenue) but do not evaluate institutional design quality. IoTeX's "DePIN sector map" classifies protocols by service type but does not distinguish architectural differences in operator base or governance structure. Borderless Capital has proposed a GeoDePIN framing emphasizing geographic coverage as a moat, but without the taxonomic precision or testable implications developed here.

In token engineering, Voshmgir [19] and the Token Engineering Commons have advanced design frameworks centered on bonding curves (mathematical functions that set token price as a function of supply) and agent-based simulation. These approaches optimize for mechanism efficiency but typically omit the normative constraints (legitimacy, fairness, contestability) that the companion framework paper argues are essential for institutional durability. The mechanism design [10], [11] literature (Hurwicz, Myerson) optimizes for incentive compatibility but does not address governance concentration or capability dimensions that DePIN uniquely raises when operators are retail individuals investing personal capital.

Lin et al. [53] provide an architectural survey formalizing the multi-layer DePIN design stack and its integration with burn-and-mint equilibrium mechanics; the present paper extends this engineering perspective by adding institutional design evaluation and longitudinal fiscal sustainability measurement.

The safety and reliability engineering community has begun analyzing DePINs as legitimate critical physical infrastructure paradigms, evaluating their resilience properties through established frameworks for distributed ownership and graceful extensibility [54].

This paper applies an institutional design lens to the specific coordination challenges of geographically distributed infrastructure, contributing a taxonomy, empirical evidence, and a practitioner-oriented playbook that existing DePIN analyses lack.

## 2. DESIGN ARCHITECTURE

The normative framework motivating these design choices (constraints derived from Kantian publicity, Rawlsian fairness, republican non-domination, and Hayekian knowledge-use) is developed in a companion study [34]. This paper retains Ostrom’s polycentric governance framework [14] as directly relevant to the geographic heterogeneity that distinguishes DePIN from other token-coordinated systems. Recent frameworks have systematically mapped Ostrom’s eight design principles to decentralized community governance [31]; the GeoDePIN taxonomy developed here extends this mapping by specifying how geographic heterogeneity creates governance challenges absent from purely digital commons.

DePIN requires an institutional architecture built around four interacting loops.

### 2.1 Design Objectives and Constraints

The mechanism design architecture implements the theoretical framework under DePIN conditions, treating protocol rules as economic institutions [26] that must satisfy four objectives simultaneously:

**Economic sustainability:** incentives must not require perpetual inflationary subsidy.

**Service reliability:** rewards must track verifiable service delivered, not merely presence.

**Political legitimacy:** rules must be publicly defensible; enforcement must be contestable.

**Social impact:** system success should include capability gains and inclusion, not only financial metrics.

Constraints include pseudonymity, sybil threats (one actor creating multiple fake identities to claim disproportionate rewards), geographic heterogeneity, oracle error (inaccurate or stale data from external data feeds), governance capture risk, and legal/regulatory uncertainty.

## 2.2 The Prime Directive: Sinks Before Faucets

The sinks-before-faucets principle, formalized as a design hypothesis in a companion study [34], motivates the following architectural constraint: A repeated fragility pattern across token systems is reward-first bootstrapping without durable demand coupling [20]. This paper formalizes the directive:

Principle 2.1 (Sinks Before Faucets): A system should deploy and verify at least one usage-linked sink before scaling emissions; burns per unit service, fee routing to treasuries that reduce circulating float.

Alshater [41] synthesizes the DePIN flywheel mechanism and the necessity of fiat-denominated pricing rails in a recent survey; the longitudinal S2R analysis developed here provides the empirical validation that survey identifies as missing from the current literature.

Violating this directive makes the system dependent on continued speculative inflows, a constitutional [4] fragility.

## 2.3 Dual-Asset Economics

DePIN systems face an adoption barrier: mainstream users prefer predictable stable pricing, while operators require upside and a credible long-run investment case. A dual-asset model, extending two-sided market logic [16], separates these needs:

Definition 2.1 (Dual-Asset Model). Let (T) be a governance/stake token and (C) be non-transferable usage credits priced in stable units [20], where:

(T) governs policy and secures operator performance (staking/slashing).

(C) is spent for service usage and obtained via (i) burning (T), or (ii) paying fiat/stables that triggers protocol buy-and-burn of (T) and mints (C).

Helium's Data Credits [2] provide an operational precedent for non-transferable usage credits spent on network usage and onboarding.

**Institutional effect:** users buy predictable service; the protocol translates demand into sink pressure; operators are compensated via governed reward policy.

## 2.4 DePIN Incentives: Pay for Service, Not Presence

DePIN's central risk is paying for "presence" (hardware online) rather than service delivered. When rewards track presence, operators deploy hardware in locations with no demand, diluting rewards for operators who serve real users and eroding network credibility [2].

Principle 2.2 (Reward Decomposition). Total operator rewards per epoch (the protocol-defined reward period, typically one day or one week)  $R_t$  should be decomposed as:

$$R_t = R_t^{base} + R_t^{usage}$$

$R_t^{base}$  maintains baseline coverage and supports frontier bootstrapping.

$R_t^{usage}$  dominates over time and pays only for SLA-verified service.

Definition 2.2 (Quality-Adjusted Usage Reward). Let  $q_{i,t}$  be a quality score for operator (i) in epoch (t), derived from SLA indicators (uptime, latency, accuracy, integrity). Let  $u_{i,t}$  be verified usage delivered (demand-weighted). Then:

$$R_{i,t}^{usage} \propto u_{i,t} \cdot q_{i,t}$$

subject to anti-gaming constraints and caps.

### **Proposed maturity schedule (normative default)**

Early: 40% base / 60% usage

Mid: 25% base / 75% usage

Mature: 10% base / 90% usage

This schedule is a policy hypothesis rather than a universal claim; it is evaluated empirically via event studies and cohort comparisons.

## **2.5 Reflexivity-Aware Emissions: The S2R Autopilot**

Token incentives are reflexive [18]: the token price affects the real cost of emissions and can induce boom-bust deployment cycles. This section proposes a rule-based stabilizer:

Principle 2.3 (Usage-Bounded Reward Budget). Reward budgets should be set in stable-value terms as a function of real usage revenue, not as an arbitrary token issuance number.

$Rev_t$  be usage-derived revenue in epoch (t) (or equivalent proxy).

$Budget_t = \alpha \cdot Rev_t$  for some governed spending fraction ( $\alpha$ ).

$Price_t^{TWAP}$  be a smoothed price measure.

$$F_t = \frac{Budget_t}{Price_t^{TWAP}}$$

$$F_t \in [(1 - \delta)F_{t-1}, (1 + \delta)F_{t-1}]$$

for ( $\delta$ ) (e.g., 5–10% per epoch). Monitoring variable: S2R

$$S2R_t = \frac{S_t}{F_t}$$

The policy aim is not a fixed number but a governance-selected band, with automatic responses when outside bounds.

**Why this is institutional (not only technical): the rule constitutes a fiscal constitution [4]: an explicit commitment to disciplined spending and transparency, consistent with transparency and signal responsiveness principles.** Proposed trigger conditions (Proposal, Normative)

These triggers extend the bounded adjustment logic above. The S2R trigger prevents runaway inflation when usage sinks fall below half of issuance, a condition Helium experienced in 2021–2022 when S2R was approximately 0.007 and HNT price crashed over 90%. The utilization trigger prevents the “coverage for coverage’s sake” failure mode by automatically shifting rewards toward demand-driven service. The uptime trigger uses staking requirements as a quality filter: if network reliability falls below acceptable thresholds, higher stakes screen out marginal operators. A companion simulation [29] is consistent with the claim adaptive emission reduces node-count deviation by 68-76% under supply shocks, but emission model choice determines the monetary regime under stress: PID-controlled protocols produce deflation (circulating supply falls 23-45%), while static protocols produce inflation (supply rises 12-47%).

Early empirical analysis of Helium’s IoT deployment documented usage patterns and infrastructure bottlenecks during the network’s pre-Solana era [42], establishing the baseline from which the S2R trajectory departs.

| Condition                                                                         | Trigger            | Proposed Action                                                       | Precedent                                                                 |
|-----------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------|
| Token burns fall below half of new issuance for N consecutive periods (S2R < 0.5) | Emission reduction | Reduce new token issuance by up to 10% per period until burns recover | Ethereum EIP-1559 [5]: base fee adjusts automatically when demand changes |



|                                                           |                      |                                                                                                      |                                                                                            |
|-----------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Less than 20% of network capacity serves paying customers | Reward rebalance     | Shift 10 percentage points of operator rewards from base (being online) to usage (serving customers) | Helium HIP-147 [9]: community vote shifted from coverage-based to service-weighted rewards |
| 95th-percentile node uptime drops below 95%               | Stake floor increase | Raise minimum operator collateral by 20%, filtering out unreliable operators                         | Filecoin [15]: storage providers post collateral proportional to capacity                  |

## 2.6 Integrity Layer: Stake, Slash, Unbonding, Appeals

In DePIN, service verification must be backed by credible enforcement [14].

Principle 2.4 (Stake-to-Earn with Contestability). Operators must post collateral and face slashing for fraud or repeated SLA failure, but enforcement must be contestable to prevent domination.

This requires four mechanisms:

**staking collateral** proportional to reward eligibility,

**slashing** for provable fraud or repeated SLA breaches,

**unbonding delays** (mandatory waiting periods before staked collateral can be withdrawn) to deter hit-and-run, and

**appeals with bounded SLA** (e.g., resolution  $\leq 14\text{--}21$  days) with published outcomes.

### Empirical evidence for integrity mechanisms

Across operating DePIN protocols, verification has emerged as the single hardest design problem. A venture capital research note from a16z Crypto states that “the hardest part of building a DePIN protocol is verification... It is the only clear way to ensure that clients receive the service” (Wuollet, a16z Crypto, March 2025). The following table summarizes how five protocols operationalize integrity:

Milionis et al. [56] formalize the verification challenge as a mechanism design problem, proving that strictly truthful location reporting requires the target node to lie within the convex hull of independent observer nodes; peripheral deployments without surrounding validators cannot be cryptographically verified, establishing formal limits on the spatial coverage claims that DePIN protocols can credibly make.

Anti-gaming measures across protocols combine technical proofs (cryptographic or hardware-based) with economic disincentives (staking and slashing), and typically require ongoing adversarial testing as attack vectors evolve. Helium’s experience illustrates the pattern: the early IoT network faced widespread spoofed locations

and self-dealing hotspots, prompting cryptographic Proof-of-Coverage challenges and a denylist that removed over 3,500 fraudulent nodes between 2021 and 2022 (Nova Labs update, August 2022). For the 5G Mobile network, Helium deployed Discovery Mapping, which uses subscriber devices to independently audit claimed coverage. Filecoin represents the strongest anti-gaming posture: cryptographic proofs make it computationally infeasible to store junk data, and slashing penalizes miners who fail to prove ongoing storage. Livepeer sends random test transcoding jobs and compares output quality against reference, an economic enforcement model (output comparison, not cryptographic proof). Hivemapper uses authenticated dashcam hardware to prevent fabricated mapping imagery. Simulation evidence from a companion study [29] reveals that slashing is not only a behavioral deterrent but a dominant monetary mechanism during the bootstrapping phase: slashing transfers 150-357M tokens from circulation to treasury across all parameterizations, while fee-funded burns contribute less than 0.05% of supply dynamics at the 5-year horizon. A downtime penalty sensitivity sweep reveals a phase transition: penalties set too low produce inflation (insufficient token removal), while penalties set too high also produce inflation through a different channel (operators are slashed out so aggressively that the network shrinks, reducing the slashing population while emission continues).

#### 2.6.1 Polycentric Governance for Heterogeneous Infrastructure

DePIN service environments vary geographically; governance should reflect this heterogeneity through polycentric structures [14]: subnet councils, regional committees, corridor working groups, each with scoped authority and budgets.

Recent frameworks have systematically mapped Ostrom's design principles to decentralized community governance [31]; the GeoDePIN taxonomy developed here extends this mapping by specifying how geographic heterogeneity creates governance challenges absent from purely digital commons.

Polycentric governance [14] supports flexibility, innovation, and resilience in complex systems.

#### 2.6.2 Bicameralism for Capital-Labor Balance

DePIN systems contain two core constituencies whose interests structurally diverge: capital investors (token holders, venture backers) who seek financial returns, and labor investors (hardware operators) who deploy personal capital and physical labor to deliver network services. Ferreras [47] argues that organizations containing both capital and labor investors are political entities requiring bicameral governance (formal representation of both constituencies with mutual veto power) to prevent unilateral capture by either. This framework maps directly onto DePIN

governance: a bicameral rule requires major economic changes to be approved by both a Token House (locked governance token holders) and an Operator/Contributor House (verified service providers).

This structure addresses two capture risks: plutocracy (capital overrides service quality, extracting unsustainable yields at operators' expense) and producer capture (operators subsidize themselves unsustainably at token holders' expense). The Optimism Collective provides the most developed implementation, with dynamic veto thresholds and mandatory cooling-off periods that prevent either house from permanently blocking the other [34]. Lijphart's [51] comparative framework formalizes the underlying trade-off: purely majoritarian governance optimizes for speed but disenfranchises minorities, while consensus mechanisms protect contestability at the cost of response time.

### 2.6.3 Constitutional Constraints

Certain parameters should be hard to change quickly: minimum transparency requirements,

mandatory appeals process and slashing bounds, data governance baseline requirements.

This maps to constitutional political economy [4] and provides a mechanism-layer counterpart to legitimacy constraints.

| Protocol   | Verification Mechanism                    | Anti-Gaming Evidence                               | Posture  |
|------------|-------------------------------------------|----------------------------------------------------|----------|
| Helium     | Proof-of-Coverage + community denylist    | 3,500+ rogue hotspots removed (2021-22)            | Strong   |
| Helium 5G  | Discovery Mapping (user device audits)    | Real-time coverage verification via subscribers    | Moderate |
| Filecoin   | Proof-of-Spacetime + Proof-of-Replication | Cryptographic enforcement; slashing for failure    | Strong   |
| Livepeer   | Random test transcoding jobs              | Output compared to reference; staked orchestrators | Moderate |
| Hivemapper | Authenticated dashcam hardware            | Hardware-enforced image provenance                 | Moderate |

## 3. METHODS

### 3.1 Data Sources and Queries

Following established case study methodology [21], the empirical analysis draws on four data source categories (Table 1): (1) On-chain data via Dune Analytics [8] queries across Ethereum, Solana, Polygon, and IoTeX chains, covering token transfers, governance voting [3], and burn transactions. (2) CoinGecko [6] API for

token price and market capitalization time series. (3) Protocol documentation including governance proposals (Helium HIPs, DIMO DIPs), whitepapers, and tokenomics specifications. (4) DeFi Llama [7] for cross-sector fee revenue benchmarks.

We measure governance concentration using the Herfindahl-Hirschman Index (HHI) computed over the distribution of governance token holdings across unique addresses, using a snapshot methodology from Dune Analytics [8] multi-chain queries as of February 2026. The governance token is the protocol's primary voting-power token; for protocols with delegated voting, the analysis uses delegated voting power distribution. We compute Gini coefficients over the same distributions. Full metric definitions and replication procedures appear in Appendix A.

### 3.2 Subsidy-to-Revenue Ratio (S2R)

The S2R measures fiscal sustainability as the ratio of demand-side spending (token burns, fee payments, credit purchases) to supply-side issuance (mining rewards, staking emissions, grants) within a defined time window. We compute monthly S2R as:  $S2R_t = \text{Burns}_t / \text{Issuance}_t$ , where Burns represents tokens permanently removed from circulation by demand-side activity, and Issuance represents new tokens entering circulation through protocol-defined reward mechanisms.

The S2R metric operationalizes the burn-to-mint equilibrium modeled as a dynamical system by Akcin et al. [37], whose control-theoretic framework predicts that stable equilibria require burn rates responsive to demand signals. Kalabic et al. [38] formalize the deflationary threshold conditions under which this equilibrium produces net supply contraction; the 34-month Helium trajectory documented in Section 5.1 provides the first empirical observation of this threshold crossing in an operating protocol. Wu and Zhu [55] demonstrate a complementary constraint: increased network utilization can paradoxically reduce token value by accelerating velocity, expanding effective supply even when burn mechanisms are active. The S2R metric sidesteps this paradox by measuring burn-to-emission ratios rather than token value accrual, but the velocity channel explains why protocols with S2R above 1.0 can nonetheless experience declining operator returns when reward recipients immediately liquidate minted tokens.

$S2R > 1.0$  indicates that a protocol is burning more tokens than it issues (fiscal parity).  $S2R < 1.0$  indicates ongoing net inflation (subsidy dependence). We report two variants: S2R (burns only) counts only permanent burns; S2R (burns + locks) additionally counts tokens locked in long-term staking or vesting contracts. We report both variants to avoid conflating irreversible and reversible demand.

S2R measures burns against algorithmic mining emissions only; vesting unlocks from team, investor, and ecosystem allocations are excluded because they represent pre-committed supply decisions rather than ongoing incentive costs that governance can adjust.

| Protocol | Burn data source                                    | Issuance data source                     | Price source |
|----------|-----------------------------------------------------|------------------------------------------|--------------|
| Helium   | Dune Analytics (Solana: action = 'burn', 149 weeks) | Dune Analytics (Solana: action = 'mint') | CoinGecko    |
| Filecoin | Filfox (burnt supply field)                         | Filfox (daily coins mined)               | CoinGecko    |
| Livepeer | N/A (no LPT burn)                                   | Subgraph (inflation parameter)           | CoinGecko    |
| Ethereum | Etherscan / Ultrasound.money                        | Etherscan (beacon chain issuance)        | CoinGecko    |

*Table 1. Data sources by protocol.*

### 3.3 Case Selection and Process Tracing

#### 3.3.1 Case Selection

Cases were selected to span variation in contestability and to include protocols with known governance disputes or redesign episodes.

#### 3.3.2 Process Tracing

The process tracing protocol examines how rule changes were proposed and justified, whether transparency and contestability mechanisms existed, how participants reacted (exit, protest, fork, apathy), and how outcomes evolved post-change. This helps distinguish correlation from mechanism and clarifies where quantitative proxies fail to capture legitimacy dynamics.

### 3.4 Ethical Considerations

All data are drawn from public sources: on-chain transactions, published governance proposals, and open protocol documentation. No private or proprietary data are used. Where protocols involve personal data collection (e.g., DIMO vehicle telemetry, CUDIS health data), we analyze governance and economic mechanisms without accessing individual-level data.

### 3.5 Sample Accounting

**To ensure transparency about the empirical scope, the following sample definitions apply throughout this paper:**

Taxonomy universe: 13 protocols classified as GeoDePIN (10, of which 2 are borderline GeoDePIN\*) or DeVIN (3). See Table 2.

Governance concentration: 8 protocols reported here (5 DePIN + 3 DeFi benchmarks) from the full 13-protocol sample. See [34].

DePIN S2R cross-section: 9 protocols with observable burn or fee data. See Table 7.

Longitudinal case: Helium, 34 monthly observations (May 2023 to February 2026). See Table 3.

Protocol profiles: 13 protocols (Helium as longitudinal case + 3 deep profiles with computed on-chain metrics + 9 thin profiles reported as a summary table).

Unit economics: 13 service domains with posted pricing data.

We treat "data-insufficient" as a finding about measurement feasibility, not as evidence of poor protocol quality. We report missingness explicitly in all tables.

## 4. THE GeoDePIN TAXONOMY

### 4.1 Definition and Design Pressures

The GeoDePIN vs. DeVIN distinction separates protocols where geographic distribution is the core defensibility mechanism from those where location is incidental. Formalizing this classification yields testable implications for governance concentration, verification complexity, and subsidy sensitivity.

### 4.2 Classification

Table 2 classifies 13 protocols along the GeoDePIN/DeVIN axis. The litmus test: if a competing network could replicate the service from a single data center without losing its core value proposition, the protocol is DeVIN; if geographic distribution across many unique locations is essential to the service, it is GeoDePIN.

Prior DePIN taxonomies separate components by architectural layer, spanning distributed ledger, cryptoeconomic, and physical infrastructure dimensions [35], or by formal decision criteria distinguishing DePIN from traditional crowdsourcing [36]. The GeoDePIN classification extends this work along a dimension neither captures: whether service value is a function of the hardware's geographic coordinates.

| Protocol   | Category | Operator Type             | Deployment Unit | Coverage Verification    |
|------------|----------|---------------------------|-----------------|--------------------------|
| Helium     | GeoDePIN | Individual hotspot owners | LoRa/5G hotspot | HIP-147 service-based    |
| Hivemapper | GeoDePIN | Individual drivers        | Dashcam         | Image authenticity + GPS |
| WeatherXM  | GeoDePIN | Individual station owners | Weather station | Cross-station validation |

| Protocol          | Category  | Operator Type                      | Deployment Unit       | Coverage Verification  |
|-------------------|-----------|------------------------------------|-----------------------|------------------------|
| DIMO              | GeoDePIN  | Individual vehicle owners          | OBD dongle / app      | Hardware attestation   |
| GEODNET           | GeoDePIN  | Individual station owners          | RTK base station      | Satellite cross-check  |
| Anyone            | GeoDePIN  | Individual relay operators         | VPN relay node        | Bandwidth verification |
| Wayru / WiFiDabba | GeoDePIN  | Community WiFi hosts               | Router / access point | Usage metering         |
| CUDIS             | GeoDePIN  | Individual wearers                 | Smart ring            | Device attestation     |
| Grass             | GeoDePIN* | Individual bandwidth sharers       | Browser extension     | Task completion        |
| UpRock            | GeoDePIN* | Individual proxy providers         | App / daemon          | Request verification   |
| Filecoin          | DeVIN     | Storage providers (institutional)  | Server rack           | Proof-of-Spacetime     |
| Livepeer          | DeVIN     | Orchestrators (semi-institutional) | GPU server            | Random test transcode  |
| Render            | DeVIN     | GPU operators                      | GPU cluster           | Frame comparison       |

Table 2. GeoDePIN vs. DeVIN classification of 13 DePIN protocols. \* = borderline; geographic distribution adds value but is not the primary defensibility mechanism.

### 4.3 Testable Implications

The GeoDePIN vs. DeVIN distinction generates testable implications beyond the descriptive classification. If the taxonomy captures real institutional differences, three patterns should hold as the sector matures: (1) GeoDePIN protocols should exhibit higher governance concentration in early stages (team/treasury dominance) but faster deconcentration schedules once operator bases diversify: the Anyone Protocol counterexample (HHI 0.034) suggests this trajectory is achievable by design; (2) GeoDePIN protocols should require earlier “service over presence” pivots in reward design, because coverage-mining traps are structurally more likely when location determines value; and (3) GeoDePIN protocols should develop stronger contestability and appeals mechanisms because hardware-bound operators face higher exit costs than software participants, making voice the primary channel for expressing dissatisfaction.

These predictions are not tested in this paper but we register them here as falsifiable claims for follow-on work.

This taxonomy clarifies why the design architecture developed in Section 2 maps onto GeoDePIN with particular force. When operators are individuals bearing personal financial risk, transparency matters more because thousands of non-technical participants need to understand reward mechanics. When coverage quality varies by geography and local conditions, polycentric governance matters more. The playbook recommendations in Section 6.1 should be read with this



distinction in mind: they are calibrated primarily to GeoDePIN coordination challenges.

Recent work on geospatially-aware consensus mechanisms [49] independently validates the geographic-coverage dimension of this taxonomy. Motepalli et al. propose non-linear consensus weighting that integrates geospatial diversity into validator influence, achieving 45% improvement in geographic decentralization across five major PoS blockchains. The GeoDePIN taxonomy extends this logic from the consensus layer to the infrastructure layer: geographic coverage is not merely a desirable validator property but the core defensibility mechanism for location-dependent service networks.

Independent industry analysis corroborates this taxonomic prediction, noting that sensor networks are 'relatively capital-light but often struggle with monetization' [28], consistent with the GeoDePIN design pressures identified above.

## 5. EMPIRICAL EVIDENCE

Claims boundary: The evidence below supports descriptive and mechanism-level claims within its observed scope. Governance concentration data represent a February 2026 snapshot. S2R trajectories are observational. Unit economics use posted prices (N = 1 per service domain). The full regression and panel pipeline remains replication-ready future work.

### 5.1 Helium: The Longitudinal Case

#### 5.1.1 S2R Trajectory

Helium's Subsidy-to-Revenue ratio rose from 0.013 in May 2023 to 1.84 in February 2026, crossing fiscal parity in October 2025 and remaining net-deflationary through the observation window. The network operates decentralized wireless infrastructure across 190+ countries through two subnetworks (IoT/LoRaWAN and Mobile/5G-CBRS) and is the primary longitudinal case in this study. Its 34-month trajectory from subsidy-dependent to revenue-dominant spans the April 2023 migration from its own L1 to Solana and a series of governance reforms, most consequentially HIP-147.

The critical reform was HIP-147 (approved Q3 2025), which shifted reward distribution from coverage-based Proof-of-Coverage to service-based metrics weighted by actual data transfer and subscriber usage. This transition directly tested whether DePIN rewards should track demonstrable service delivery rather than mere infrastructure presence.



*Figure 1. Helium Subsidy-to-Revenue (S2R) ratio trajectory, May 2023 – February 2026 (burns only, 34 monthly snapshots from Dune Analytics [8] on-chain data). Dashed line at  $S2R = 1.0$  indicates fiscal parity where token burns offset new issuance. Three regimes are visible: subsidy-dependent ( $S2R < 0.15$ , 2023–early 2025), transitional (mid-2025), and net-deflationary ( $S2R > 1.0$ , October 2025 onward). HIP-147 (September 2025) marks the shift from coverage-based to service-weighted rewards. See Table 3 for selected data points.*

Three fiscal regimes emerge across the 34-month S2R trajectory (Table 3): (1) post-migration baseline ( $S2R$  0.01 to 0.04, May 2023 to early 2024), when burns were minimal and issuance dominated; (2) demand ramp ( $S2R$  0.15 to 0.67, mid-2024), as Mobile subscriber growth and IoT data transfer increased burn volume; (3) fiscal parity and beyond ( $S2R$  1.03 to 1.84, late 2025 to February 2026), when monthly burns consistently exceeded issuance.

Helium became the first GeoDePIN protocol where demand-side spending fully offsets supply-side inflation. DIMO subsequently achieved  $S2R = 4.24$ , making it the second protocol to demonstrate fiscal parity through demand-coupled burns. A pricing caveat applies to Helium's December 2025 peak: WiFi offload traffic is priced at \$0.10-0.25 per GB versus \$0.50 for CBRs, and the annualized burn figure reflects a temporary Mobile burn allocation subsequently adjusted, so the S2R spike overstates steady-state equilibrium by an estimated 2-5x.

The fiscal vulnerability of the early S2R regime is illustrated by the San Jose broadband pilot (2021–2022). The city deployed 20 Helium hotspots targeting 1,300 low-income households for broadband subsidies funded by HNT mining revenue; only 86 households were ultimately served after token price volatility destroyed the fiat-denominated value of mining rewards [50]. The failure occurred during the post-migration baseline period ( $S2R$  0.01–0.04), when no demand-coupled burns anchored the token's real value. The case provides causal evidence that token-subsidized infrastructure programs fail when the sinks-before-faucets principle is violated: without demand-coupled burns, the subsidy's real value is entirely determined by speculative token dynamics rather than network utility.

Independent industry data corroborate this trajectory: by December 2025, Helium's annualized Data Credit burn had reached a \$20.9M run-rate (trailing three-month annualized, October–December 2025), representing an eightfold increase from the prior year even as HNT's token price declined 77% over the same period [28] (the companion paper [34] cites a smaller \$18.3M figure for Helium's annualized gross revenue at Q3 2025; the two measures differ because burn run-rate captures consumption-side activity while revenue captures the cash side, over different observation windows). This revenue-price decoupling is consistent with S2R capturing demand-side fundamentals independently of speculative token dynamics.

| Period   | S2R (burns only) | HNT Burned | HNT Issued | Context                                   |
|----------|------------------|------------|------------|-------------------------------------------|
| May 2023 | 0.013            | 37,204     | 2,877,911  | Post-migration baseline, IoT only         |
| Sep 2023 | 0.119            | 138,660    | 1,170,115  | Early 5G traction, first meaningful burns |
| Jan 2024 | 0.052            | 75,592     | 1,458,899  | Steady-state pre-carrier-burn             |
| Jun 2024 | 0.024            | 28,152     | 1,162,966  | Mid-cycle trough                          |
| Nov 2024 | 0.137            | 159,517    | 1,163,406  | Spike month (possible large DC purchase)  |
| Mar 2025 | 0.107            | 200,362    | 1,876,522  | Pre carrier-burn, growing demand          |
| Jun 2025 | 0.376            | 708,897    | 1,883,509  | Full carrier revenue burn begins          |
| Aug 2025 | 0.869            | 532,886    | 613,510    | Post-halving: issuance drops ~58%         |
| Sep 2025 | 0.587            | 456,181    | 776,712    | HIP-147 passes; S2R pulls back            |
| Oct 2025 | 1.570            | 975,288    | 621,370    | Burns exceed issuance for first time      |
| Dec 2025 | 1.552            | 1,205,193  | 776,712    | Sustained net-deflationary                |
| Feb 2026 | 1.841            | 1,144,240  | 621,370    | Burns at ~1.8x issuance                   |

*Table 3. Helium monthly S2R trajectory (May 2023 to February 2026, N = 34).*

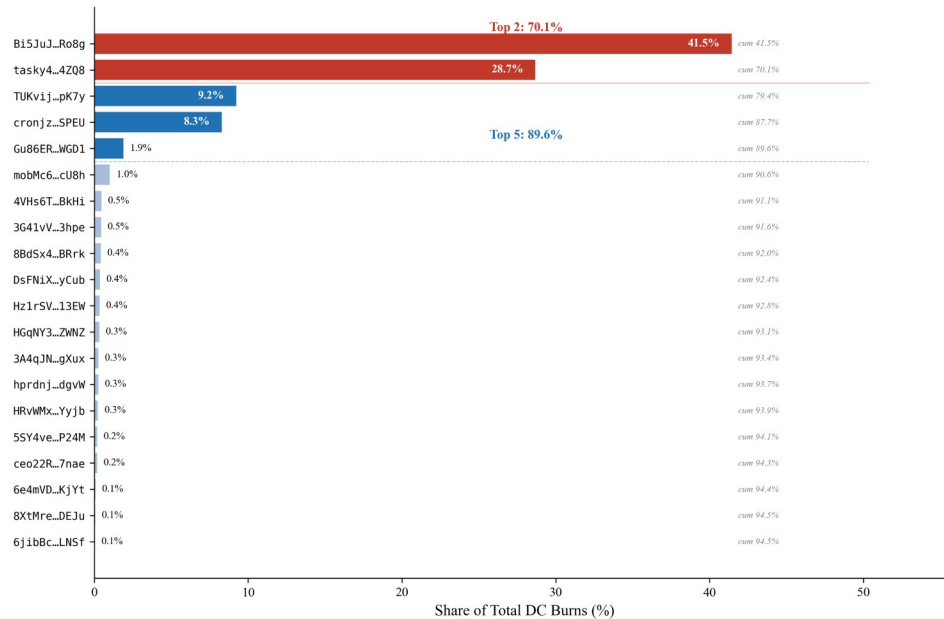
### 5.1.2 Who Burns? Demand Concentration

Dune Analytics [8] validation reveals that HNT burns are highly concentrated. The top 5 burn-signing addresses account for approximately 90% of total HNT burned across the 34-month observation window. The single largest address (Bi5JuJD...) processed 41% of all burns (3.25 million HNT) in only 180 transactions, consistent with institutional or treasury Data Credit purchases rather than grassroots operator demand. The second-largest signer (tasky4vv..., 2.25 million HNT in 43,600+ transactions since November 2025) is an automated purchasing bot, serving a carrier partner. A burn-signer HHI of approximately 0.27 places Helium’s demand side in the “moderately concentrated” range by the same antitrust thresholds applied to governance tokens (Section 5.5).

Resource Dependence Theory [43] predicts that this concentration transfers strategic leverage to the concentrated buyers independent of formal governance arrangements. If the dominant burn signer ceased purchasing Data Credits, Helium’s S2R would fall from 1.84 to approximately 0.62 within one month, returning the protocol to subsidy dependence regardless of how governance tokens are distributed. The concentration risk is compounded by roster instability: the dominant February 2026 signer (66.4% of monthly burns) was not active before November 2025, having displaced the prior leading address, which fell from 45.4% of cumulative burns to under 1% of February volume. Governance concentration and demand concentration are therefore independent failure modes: the first determines who controls protocol rules, the second determines who controls protocol revenue.

Industrial organization theory formalizes this demand-concentration risk as protocol oligopsony [26]: when a small number of governance token holders unilaterally control emission rates and reward parameters, they function as

monopsonist buyers of operator infrastructure services. When buyer power exceeds the efficient threshold, the resulting input-price markdown causes operator exit and network contraction, the supply-side abandonment pattern documented above and predicted by the health dashboard’s demand-concentration guardrails (Section 6.3).



Source: Dune Analytics Q6942532 (Solana, Jan-Mar 2026). N = 20 shown (top-20 of 20 total burn signers).

**Figure 2. Helium "Who Burns?": top burn addresses by Data Credit volume (2024).**  
*Two addresses account for 70.1% of total burns (HHI = 0.27).*

This concentration does not negate the S2R mechanism; it clarifies what S2R measures. Aggregate DC demand is genuinely growing (burns consistently exceed issuance since October 2025) and the deflationary pressure is real regardless of whether it originates from millions of retail users or a few carrier contracts. S2R is therefore best understood as a fiscal sustainability indicator (demand coupling and sink efficacy), not as a distributed adoption metric. The distribution of demand is a separate property that should be monitored via burn-signer concentration; if Top-5 signer share remains above 80% for an extended period, the network faces counterparty concentration risk that the health dashboard (Section 6.3) should flag.

This distinction also refines the service-over-presence principle: S2R supports the claim that demand-coupled sinks work as a fiscal mechanism; whether that demand reflects broad-based adoption or concentrated institutional purchasing is orthogonal to the sink design question and requires its own measurement layer.

Hypothesis mapping: The strongest single case for the service-over-presence principle. The S2R trajectory from 0.013 post-migration (May 2023) to 1.84 in February 2026 (with burns exceeding issuance since October 2025) demonstrates that shifting from presence-based to service-based rewards can transform a network from deeply inflationary to structurally deflationary. The August 2025 halving and the full carrier revenue burn policy are the two causal drivers; the on-chain data cannot separate their contributions without a formal event study, but the pre-halving jump to 0.38 in June 2025 (carrier burn only, no halving) supports that demand-side burns alone moved S2R substantially. The trajectory also supports Hypotheses 3.3 (sinks before faucets: the burn mechanism was necessary but insufficient without demand) and 3.5 (polycentric governance enabled the pivot via community-voted HIPs rather than centralized decree). The companion simulation [29] independently corroborates the demand-concentration risk: even with a 30% protocol burn rate, burns are negligible relative to emission when aggregate fee revenue is low, meaning the entire burn-mint sustainability thesis depends on achieving the revenue concentration that Helium's top-5 Data Credit signers provide.

Rad et al. [58] independently corroborate this trajectory, demonstrating empirically that long-term token valuation in Helium's burn-and-mint ecosystem depends on actual data transmission metrics rather than hardware deployment scale alone.

### 5.1.3 Hypothesis Mapping

The Helium case provides the strongest single-protocol evidence for two design principles: sinks before faucets, demonstrated by the S2R trajectory crossing parity only after demand-coupled burns were deployed; and service over presence, demonstrated by HIP-147's measurable fiscal impact. The demand concentration finding introduces a caution not anticipated by the companion framework: aggregate S2R can mask counterparty risk.

## 5.2 Deep Profiles

### 5.2.1 DIMO

DIMO operates a decentralized connected vehicle data marketplace on Polygon. The unit of service is vehicle telemetry data (OBD-II diagnostics, GPS, and driving behavior) shared by car owners who install DIMO hardware dongles or connect via software integrations. Data consumers (insurers, fleet managers, researchers) purchase access through the DIMO Developer License, which requires burning DIMO tokens. As of February 2026, DIMO has a market capitalization of \$4.5M with 424 million circulating tokens out of 1 billion total supply (CoinGecko [6]).

DIMO's sink architecture features a license burn mechanism: developers must burn DIMO tokens to mint a Developer License NFT for API access to vehicle data. This creates a demand-coupled burn analogous to Helium's Data Credit model. On-chain data from Dune Analytics [8] reveals an S2R of 4.24 for the most recent monthly window (computed as total token burns divided by total issuance in that period), indicating that burns significantly exceed minting, though this reflects a period of elevated developer license purchases against low emission rates rather than sustained equilibrium.

After excluding the DIMO Foundation address (verified via Blockscout, holding approximately 55% of supply with no on-chain governance activity), DIMO's governance HHI drops from 0.305 to 0.038 [34]. Post-exclusion, DIMO is among the least concentrated DePIN governance tokens in the 40-protocol cross-section. The corrected value strengthens the observation that DIMO's high S2R (4.24) coexists with distributed governance.

Verification relies on hardware device attestation (DIMO Macaron dongle) and software integrations with OEM APIs. Data authenticity is ensured by signed device certificates. The verification posture is moderate: hardware-based provenance for directly connected vehicles, but dependent on third-party APIs for software-only integrations.

DIMO's sustainability signal is early-positive. The license burn mechanism creates a direct pathway from data demand to token value, and the S2R above unity suggests meaningful demand-side activity. However, the small market capitalization (\$4.5M) and high governance concentration suggest the protocol is in its growth phase. The unit economics comparison with Web2 vehicle data providers (Otonomo, Wejo at \$0.50–3.00/vehicle/month) is not directly comparable because DIMO users share data voluntarily for token rewards rather than paying a subscription. The novel data ownership model (where drivers control and monetize their own vehicle data) represents a capability expansion for drivers who previously lacked access to their own data.

Design lesson: DIMO illustrates the sink-before-scale principle (Section 2.2): the Developer License burn was operational before aggressive hardware distribution, ensuring token demand scaled with genuine data consumption. Pre-exclusion governance concentration (HHI 0.305, reduced to 0.038 post-exclusion) represents the structural risk that bicameral governance (Section 6) addresses.

### 5.2.2 Hivemapper

Hivemapper operates a decentralized street-level mapping network on Solana. The unit of service is freshness-verified map imagery, generated by contributors who

install authenticated dashcams in their vehicles. Mapping clients consume map data by purchasing Map Credits, which requires burning HONEY tokens. As of February 2026, Hivemapper has a market capitalization of \$20.8M with 5.6 billion HONEY circulating (CoinGecko [6]). The network has mapped portions of roads in 100+ countries, though coverage density varies significantly by region.

Hivemapper's sink architecture implements a burn-and-mint model (users burn tokens to purchase service credits; operators earn newly minted tokens for providing service) similar to Helium's DC system. Map Credits are purchased with HONEY (burned), and contributors earn HONEY for submitting verified imagery. MIP-7 introduced tiered pricing based on map freshness and coverage. Dune Analytics queries (February 2026) yield  $S2R = 0.183$  (1.9M HONEY burned against 10.4M minted), confirming that the burn mechanism is active but that Hivemapper remains subsidy-dependent: burns offset less than one-fifth of issuance. The mechanism design is structurally similar to Helium's DC burn model, and the measurable S2R places Hivemapper in the growth-phase cohort between subsidy-dependent protocols (S2R near zero) and fiscally self-sustaining pairs (Helium at 1.84, DIMO at 4.24).

Governance concentration data are unavailable for Hivemapper: the HONEY token on Solana is not indexed by the Dune multi-chain governance query used for the cross-section (Section 5.5). This data gap is itself informative: Solana-native tokens remain less transparent to standardized governance analysis than EVM-chain tokens, creating a measurement asymmetry that systematically excludes Solana DePIN protocols from comparative governance research. Hivemapper's governance operates through a foundation-led improvement proposal process (MIPs), with MIP-7 (tiered map freshness pricing) representing the most significant economic parameter change to date.

Hivemapper's sustainability signal is early-stage but directionally positive. The map credit burn mechanism creates a structural demand sink, and the network's coverage across 100+ countries represents a genuine data asset with freshness advantages over static satellite imagery. Unit economics show a 29% discount versus Google Maps API pricing (Table 5), with Medium subsidy risk: dashcam operators are compensated primarily through HONEY emissions, but the \$300–600 hardware investment and fuel costs create natural filtering against purely mercenary participation.

Design lesson: Hivemapper demonstrates the verification-as-moat pattern identified in the design architecture (Section 2.6). Authenticated dashcam hardware enforces image provenance at the physical layer, making coverage fraud economically

irrational without manufacturing counterfeit devices. This hardware-enforced integrity contrasts with Helium's early experience, where software-only Proof-of-Coverage was vulnerable to widespread spoofing. The Hivemapper case supports the playbook recommendation that DePIN protocols should invest in hardware attestation before scaling operator rewards; integrity built in from launch. The partial Solana data gap also motivates the health dashboard recommendation (Section 6.3) that protocols publish standardized burn and governance metrics regardless of chain.

### 5.2.3 WeatherXM

WeatherXM's sink architecture combines a data subscription burn (WXM burned for API access) with emission rewards to station operators. The token launched on Arbitrum mainnet in May 2024 with a 100M total supply and 10-year distribution schedule. Governance concentration is the highest in the expanded cross-section (Section 5.5): HHI 0.593 with a single address holding 76.4% of tokens. This extreme concentration reflects the recency of the token launch and team/treasury holdings that have not yet been distributed. The 52M token allocation to weather station owners over 10 years suggests concentration should decrease as the distribution schedule progresses.

Verification relies on station hardware attestation and cross-validation against neighboring stations and satellite data. WeatherXM uses a reward algorithm that penalizes stations reporting data inconsistent with nearby stations or known weather patterns. The anti-gaming posture is moderate: hardware-based provenance plus statistical cross-validation, but sophisticated spoofing of meteorological data remains theoretically possible.

WeatherXM's sustainability signal is early-stage but structurally sound. The data marketplace creates a direct link between weather data demand and token burns. Unit economics suggest a 10x discount versus Web2 weather API providers (Tomorrow.io at \$0.05–0.20/API call vs. WeatherXM at ~\$0.01/call), with Medium subsidy risk. Station operators are compensated primarily through token emissions, but the hardware investment (\$200–400) creates natural operator commitment. The approximately 9,800-station network across 96 countries provides a genuine data asset: hyperlocal weather data at station density that exceeds many national meteorological services in covered areas.

Design lesson: WeatherXM exemplifies the governance concentration paradox identified in the cross-section (Section 5.5). Despite a structurally sound sink mechanism and genuine data utility, the protocol exhibits the highest HHI in the sample (0.593) because token distribution has barely begun. The 10-year

distribution schedule to station owners should progressively dilute this concentration, but the current snapshot represents a period where governance is effectively centralized regardless of on-paper decentralization claims. This supports the playbook's recommendation (Section 6.1) that token distribution timelines should front-load operator allocations to establish governance legitimacy before major protocol decisions are made.

#### 5.2.4 IoTeX

IoTeX operates a purpose-built Layer-1 blockchain optimized for Internet of Things devices and DePIN applications. The platform [16] sells IoT middleware (W3bstream) that enables devices to generate verifiable proofs of real-world activity, bridging physical hardware events to on-chain token rewards. Device manufacturers and DePIN projects build on IoTeX infrastructure, paying IOTX gas fees for transactions. With a market capitalization of \$51M and 9.4 billion circulating tokens, IoTeX is among the most mature DePIN infrastructure protocols.

IoTeX employs an EIP-1559 [5]-style gas fee burn mechanism on its native L1 chain. Transaction fee [17]s are partially burned, creating a usage-proportional sink. Staking locks reduce effective circulating supply; approximately 40% of IOTX is staked with network delegates. The governance concentration data reveals extreme concentration: HHI 0.388 with 61.3% of top-1000 holder tokens in a single address and 91.7% in the top 10. This reflects staking pool aggregation and team treasury holdings, not broad retail distribution.

Verification relies on W3bstream's trusted execution environment, which is consistent with IoT device data before issuing on-chain proofs. Devices register cryptographic identities and submit signed data streams. The integrity layer is moderate: hardware-based attestation plus economic stake, but without the cryptographic proof-of-service rigor of Filecoin's Proof-of-Spacetime.

IoTeX's sustainability signal is mixed. Gas fee burns create a direct usage-to-token sink, but the absolute burn volume is small relative to staking emissions. The protocol's competitive positioning as DePIN infrastructure middleware is distinctive: it provides infrastructure middleware, competing with general-purpose L1s and IoT cloud platforms (AWS IoT Core). The governance concentration finding (HHI 0.388) indicates that despite IoTeX's maturity, token distribution [12] remains highly concentrated on the Ethereum representation, reflecting cross-chain bridge dynamics rather than underlying governance structure.

Design lesson: IoTeX illustrates the platform-layer measurement challenge: S2R cannot be computed from a single burn stream for infrastructure middleware; value accrues across heterogeneous DePIN applications built on the chain. Governance



concentration (HHI 0.388) may reflect Ethereum-side token bridging rather than L1 governance participation. Platform-layer protocols require composite sustainability indicators rather than single-stream S2R.

### 5.2.5 GEODNET

GEODNET operates a decentralized network of Real-Time Kinematic (RTK) base stations on Solana (primary, since GIP7) and Polygon (original chain). The unit of service is centimeter-level GNSS positioning corrections, consumed by drones, autonomous vehicles, agricultural equipment, and construction machinery. Enterprise clients (50+, including Quectel and DroneDeploy) purchase subscriptions whose revenue funds a buy-and-burn mechanism. As of March 2026, GEODNET has a market capitalization of \$59M with 425 million circulating tokens out of 968 million total supply (CoinGecko [6]), a network of 20,000+ stations across 150+ countries, and annualized recurring revenue of \$7.3M (December 2025).

GEODNET's sink architecture implements a subscription-funded buy-and-burn: 80% of Foundation revenue (after a 50/50 share with third-party resellers on indirect channels) purchases GEOD on the open market and burns it, creating a demand-coupled sink whose effective burn rate ranges from 40% of gross revenue (reseller channels) to 80% (direct sales). Dune Analytics [8] queries on Solana SPL token burn instructions reveal 15.1 million GEOD burned since the Solana migration (September 2024), with monthly burns ranging from 163,000 GEOD (October 2024, early migration) to 2.17 million GEOD (January 2025, peak). February 2026 burns totaled 1.34 million GEOD, yielding the S2R of 0.62 reported in Table 4. Annual token halving (June 30 each year) progressively reduces issuance: post-July 2025 halving, triple-band miners earn 12 GEOD/day, down from 24. The combination of growing burns and declining issuance targets net-positive tokenomics (burns exceeding issuance) by 2026–2027. A demand-concentration analysis (the “Who Burns?” methodology from Section 5.1.2) reveals that GEODNET's burn demand is substantially less concentrated than Helium's: the top-5 burn signers account for 40.4% of total burns (burn-signer HHI 0.055), compared to Helium's top-5 at approximately 90% (burn-signer HHI 0.27). The top-1 GEODNET signer holds 9.4% of burns versus Helium's 43%. This difference is structurally informative: GEODNET's subscription model distributes burn demand across dozens of enterprise clients, whereas Helium's carrier-contract model concentrates burns in a few large institutional purchasers. The finding directly supports the health dashboard's recommendation (Section 6.3) that demand concentration should be monitored alongside S2R: GEODNET's lower burn concentration suggests greater demand-side resilience to single-buyer departure.

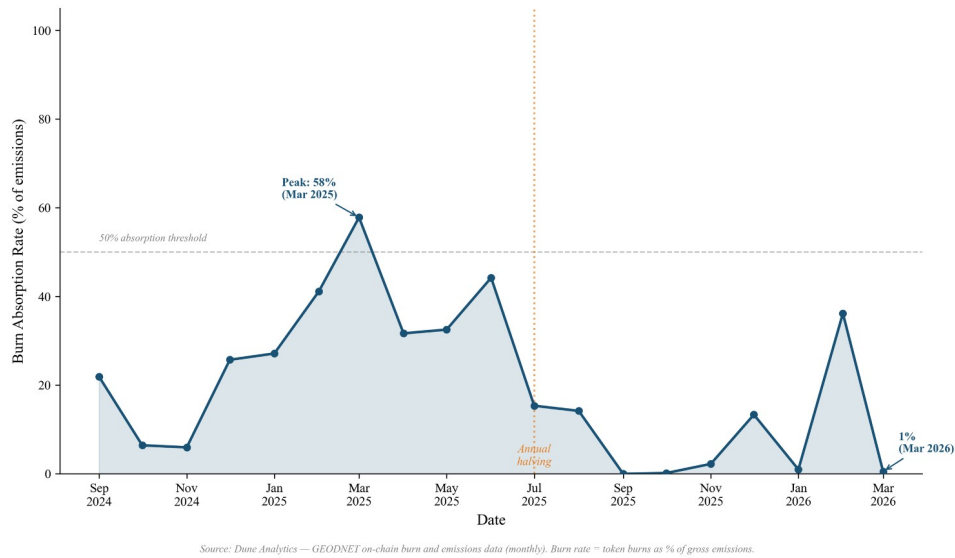
Governance concentration on the Polygon-side token distribution yields HHI 0.14 with the team wallet holding 25.3% of non-burned supply and the Wormhole NTT bridge contract (aggregating Solana holders) holding 21.1%. Mining reserves (13.7%) and ecosystem reserves (6.4%) account for additional protocol-controlled concentration. The measurement carries the same cross-chain caveat as IoTeX (Section 5.2.4): 204 million GEOD (21% of total supply) has been bridged to Solana, where it is held by thousands of individual wallets not individually resolved in the Polygon snapshot. The Polygon-side HHI therefore overstates true governance concentration by treating the bridge contract as a single entity. Governance operates through GEODNET Improvement Proposals (GIPs), with GIP3 (Solana migration) and GIP7 (performance staking and Solana-primary designation) representing the most significant economic parameter changes to date.

Verification relies on a multi-layer integrity stack: Proof-of-Location (PoL) validates station geographic coordinates against satellite reference data, Proof-of-Accuracy (PoA) evaluates RTK correction quality against known baselines, and BLS signatures provide Byzantine Fault Tolerance across the station network. GIP7 introduced performance staking with quality scoring thresholds (98–99.9% uptime required), creating economic incentives aligned with data reliability. Stations that fail accuracy or uptime thresholds receive reduced rewards, a proportional sanction consistent with the integrity loop architecture (Section 2.6). The first station deployed in a geographic hex earns a Location NFT, creating a non-fungible claim on spatial priority.

GEODNET's sustainability signal is the strongest in the growth-phase tier. The S2R accelerated from approximately 0.15 in mid-2024 to 0.62 by early 2026 [32], exceeding Hivemapper (0.183) and approaching the trajectory that Helium followed before crossing fiscal parity. By early 2026, the network spanned 5,802 cities with median station earnings of \$47.63 per month [32], and weekly GEOD burns had reached 1.06 million tokens. Three structural factors support continued S2R improvement: (1) the 80% burn rate converts revenue directly to token removal; (2) annual halving reduces the issuance denominator; (3) ARR growth of 18% month-over-month (December 2025) increases the burn numerator. Unit economics show RTK positioning at \$1–5/station/month versus Trimble/Hexagon incumbents at \$50–100/station/month (Table 5), a 10–80x cost advantage with Medium subsidy risk. The enterprise client base (50+ organizations) and the non-discretionary nature of precision positioning for autonomous systems suggest demand resilience beyond speculative token use.

The trajectory is not monotonic: accelerating vesting unlocks in Q2 2025 (monthly unlocks rose 60% from 23 million to 30 million tokens as team and investor

allocations vested) temporarily overwhelmed subscription burn growth, collapsing the absorption rate despite a scheduled June 30 halving that cut mining rewards by 50% [45]. By January 2026, subscription growth had partially restored absorption to 62%. This pattern exposes a supply-side risk the S2R metric does not capture: vesting-driven dilution operates independently of emission schedules and can overwhelm organic demand even when the protocol’s mining rewards are contracting.



*Figure 3. GEODNET cumulative burn trajectory (2022–2025). Dashed line marks the BME crossover ( $S2R = 1.0$ ); shaded region indicates net-deflationary quarters.*

By early 2026, the network spanned 5,802 cities with median station earnings of \$47.63 per month [32], and weekly GEOD burns had reached 1.06 million tokens.

Design lesson: GEODNET introduces the SuperHex staking model, a spatial incentive mechanism absent from all other protocols in the 13-protocol sample. Under SuperHex, token holders stake GEOD to designate high-value geographic hexagons (H3 cells) before stations are deployed, creating a market signal for where coverage is most valuable. Capital flows to frontier regions before hardware does, directly implementing the “spatially intelligent incentives” design pressure identified in Section 1.2. This contrasts with Helium’s early experience, where uniform global reward rates systematically over-rewarded saturated urban areas. SuperHex converts the abstract design principle (location-weighted rewards) into a staking-based mechanism where the community collectively allocates geographic priority. The burn-concentration finding extends the design lesson: GEODNET’s subscription-based burn model produces a burn-signer HHI (0.055) that is approximately one-fifth of Helium’s (0.27), suggesting that subscription-based demand architectures

may be structurally more resilient than carrier-contract models, a testable hypothesis for the cross-protocol program outlined in Section 7.4.

### 5.3 Comparative Protocol Summary

We report the remaining nine protocols as a structured summary (Table 4). For protocols labeled "data-insufficient," on-chain observability is missing or the data schema prevents S2R computation. Where we mark "data-insufficient," this is itself a finding about protocol legibility: the absence of instrumentable data is evidence of measurement immaturity and governance transparency deficit.

| Protocol        | Service          | Sink Type         | S2R    | Gov HHI | Subsidy Risk | Key Finding                                                     |
|-----------------|------------------|-------------------|--------|---------|--------------|-----------------------------------------------------------------|
| Filecoin        | Storage          | Collateral + burn | ~0.045 | –       | Medium       | Strong verification; low S2R (0.045)                            |
| Livepeer        | Video transcode  | Fee routing       | ~0     | –       | Low          | Orchestrator competition drives cost advantage                  |
| Hivemapper      | Street mapping   | Map credit burn   | 0.183  | –       | Medium       | Freshness advantage; S2R 0.183 (Dune Feb 2026)                  |
| GEODNET         | RTK positioning  | Subscription burn | 0.62   | 0.14*   | Medium       | S2R 0.62; burn Top-5 = 40.4% (Section 5.2.5)                    |
| Anyone          | VPN/relay        | Token burn        | ~0     | 0.034   | High         | Most diffuse governance in DePIN sample                         |
| Grass           | AI training data | None yet          | ~0     | 0.119   | Critical     | No burn mechanism; Gini 0.96; AI positioning                    |
| UpRock          | Bandwidth        | None yet          | ~0     | –       | Critical     | Pre-revenue; data-insufficient                                  |
| Wayru/WiFiDabba | WiFi             | Micro-payments    | –      | –       | Medium       | Last-mile focus; data-insufficient                              |
| CUDIS           | Health data      | Data marketplace  | –      | –       | High         | Privacy-critical health data; regulatory compliance constraints |
| Render          | GPU compute      | Fee burn          | –      | –       | Low          | DeVIN benchmark; institutional operators                        |

Table 4. Comparative protocol summary. –, " = not measured; "N/A" = not applicable; "data-insufficient" = on-chain observability missing or schema prevents computation.

A three-tier sustainability hierarchy emerges from the comparative summary (Table 4). At the top, Helium (S2R 1.84) and DIMO (S2R 4.24) have crossed fiscal parity, with burns exceeding issuance. In a growth-phase middle tier, GEODNET (S2R 0.62) and Hivemapper (S2R 0.183) show confirmed on-chain burn activity but remain subsidy-dependent. The remaining protocols cluster near S2R zero, the default condition for DePIN. Two additional findings sharpen the picture. Governance data are missing for protocols not on chains with transparent voting records, meaning governance risk is unmeasured rather than absent. And the subsidy risk column reveals a bimodal distribution: protocols with established demand sinks face medium risk, while those without any burn mechanism face critical risk regardless of positioning narrative. A companion simulation [29] calibrated to Helium parameters shows that a comparable protocol does not approach S2R parity within

5 years under realistic demand assumptions (peak S2R = 0.008), consistent with Helium's 7+ year journey to fiscal sustainability. This suggests that the S2R crossing is a late-stage achievement and that protocols should design emission budgets for a multi-year subsidy-dependent phase rather than expecting rapid crossover.

#### 5.4 Unit Economics: DePIN vs. Web2

DePIN services are already price-competitive with Web2 incumbents across multiple domains, though subsidy risk varies (Table 5). The comparison draws on the Deep Research evidence base, verified against DeFi Llama [7] fee data. Each row includes a subsidy risk assessment, since DePIN cost advantages may partially reflect token emission subsidies rather than structural efficiency gains.

Empirical performance analysis indicates that DePIN serverless architectures can match or exceed centralized cloud providers in computational efficiency while reducing reliance on centralized trust assumptions [57], consistent with the unit economics advantages documented in Table 5.

Field deployment of decentralized LoRaWAN environmental sensors in Nairobi achieved a 99.6% reduction in operational networking costs relative to legacy 4G telecom alternatives [59], the largest empirical cost advantage documented for DePIN infrastructure in a low-resource urban environment.

Sources: Messari State of Helium Q3 2025; Livepeer Studio press release, May 2024; Filecoin market data via Filfox; Hivemapper MIP-7 pricing; Deep Research unit economics tables. All prices USD, 2024–2025 data.

*Subsidy risk scoring:* **Low** = service priced above marginal operator cost, sustainable without token emissions. **Medium** = price below operator cost but viable if emissions taper 50%. **High** = price significantly below operator cost, dependent on ongoing token emissions. Helium Mobile's \$30/month plan is rated High because hotspot operators are compensated primarily through HNT rewards rather than subscriber fees; if rewards decline, either the plan price must rise or operators must also be subscribers (increasing owner-user overlap). Livepeer is rated Medium because orchestrators earn ETH fees directly from clients, but LPT staking rewards supplement income. Filecoin is rated Low-Medium because miners bid competitively and collateral requirements filter for economically viable operators.

*Measurement note:* All prices observed February 2026 from public protocol documentation. These are point-in-time posted-price comparisons; results may differ across transaction volumes or time periods.

Across all five domains, DePIN pricing undercuts Web2 incumbents by 1.4–80x, though the subsidy caveat applies to three of five: the cost advantage may narrow as token emissions decline.

## DeFi Llama [7] Revenue Comparison: DePIN vs DeFi Protocols

For context, DePIN protocol revenues remain orders of magnitude below mature DeFi protocols. The following table reports average daily fee revenue from DeFi Llama [7] (30-day average, February 2026):

Source: DeFi Llama [7] API (February 2026). Revenue = fees accruing to protocol treasury or token holders (distinct from total fees paid by users).

DePIN fee revenue is 100–1000x smaller than DeFi fee revenue. This reflects the sector’s maturity: DePIN networks are infrastructure plays in early growth phases, while DeFi protocols have achieved product-market fit with billions in daily volume. The comparison is not a failure signal but a stage-of-life indicator. Independent corroboration comes from Messari’s 2025 sector review [28]: Helium’s onchain revenue grew approximately eightfold while HNT declined 77% over the same period, demonstrating the revenue-price decoupling that S2R is designed to capture. We evaluate mechanism design propositions on trajectory (is S2R rising? is utilization growing?) rather than absolute revenue level.

| Domain          | DePIN Network   | DePIN Price           | Web2 Comparator  | Web2 Price           | Discount | Subsidy Risk |
|-----------------|-----------------|-----------------------|------------------|----------------------|----------|--------------|
| IoT (LoRaWAN)   | Helium IoT      | \$0.00001/<br>packet  | Senet            | ~\$0.0008/<br>packet | ~80x     | Medium       |
| Mobile 5G       | Helium Mobile   | \$30/mo<br>unlimited  | T-Mobile         | \$70/mo<br>unlimited | ~57%     | High         |
| Video transcode | Livepeer        | \$0.005/min           | AWS MediaConvert | \$0.024/min          | 50-80%   | Medium       |
| Storage         | Filecoin        | ~\$0.01/GB-<br>month  | AWS S3           | \$0.023/GB-<br>month | >50%     | Low-Medium   |
| Mapping         | Hivemapper      | ~\$5/1000<br>tiles    | Google Maps API  | \$7/1000<br>loads    | ~29%     | Medium       |
| Bandwidth/proxy | UpRock          | ~\$1.50/GB            | Bright Data      | \$12.75/GB           | 8.5x     | High         |
| AI data         | Grass           | Indirect<br>(tokens)  | Scale AI         | \$25-40/hr           | N/A      | High         |
| Vehicle data    | DIMO            | Free (earn<br>tokens) | Otonomo          | \$0.50-3/<br>veh/mo  | N/A      | High         |
| Weather data    | WeatherXM       | ~\$0.01/API<br>call   | Tomorrow.io      | \$0.05-<br>0.20/call | 10x      | Medium       |
| VPN relay       | Anyone          | Free relay<br>(earn)  | NordVPN          | \$3.49-<br>8.32/mo   | N/A      | High         |
| Community WiFi  | Wayru/WiFiDabba | ~\$2-5/mo             | ISP retail       | \$15-30/mo           | 6.4x     | Med-High     |
| Wearable health | CUDIS           | ~\$69 ring            | Oura Ring        | \$299 +<br>\$6/mo    | 4.3x     | High         |
| IoT infra (L1)  | IoTeX           | ~\$0.01/tx            | AWS IoT Core     | \$1/M msgs           | N/A      | Medium       |

*Table 5. DePIN vs. Web2 unit economics across 13 service domains.*

## 5.5 Governance Concentration

Governance concentration across DePIN and DeFi protocols is analyzed in a companion cross-sectional study using Blockscout-verified exclusion methodology across 40 protocols [34]. That study documents that holding-based concentration is higher in DePIN than DeFi after protocol-controlled address exclusion (Mann-Whitney  $p = 0.031$ , Cohen's  $d = 0.96$ ), though the result is sensitive to sample composition. Delegation amplifies DePIN concentration 3 to 6 times above holding-based levels, while two infrastructure protocols diverge: Optimism 0.79x and Arbitrum 4.3x. The present paper uses protocol-specific governance observations as inputs to the design playbook below.

## 6. DESIGN PLAYBOOK

Every design recommendation below ties to an empirical or architectural anchor from Section 5; recommendations without such grounding appear in Supplementary Material S1 (Practitioner Brief).

### 6.1 Design Implications

The data reveal a sector where most protocols are simultaneously governance-concentrated and subsidy-dependent. Only four protocols show confirmed burn activity above zero: Helium (S2R 1.84), DIMO (4.24), GEODNET (0.62), and Hivemapper (0.183); Helium and DIMO have crossed fiscal parity ( $S2R \geq 1.0$ ), with GEODNET on an accelerating trajectory (Table 7). The normative framework motivating broader tokenomics design choices is developed in a companion study [34]. The following implications derive from this paper's own empirical evidence.

**Demand concentration as independent health metric.** The “Who Burns?” analysis (Section 5.1.2) reveals that high S2R can coexist with concentrated demand, creating single-buyer fragility. Burns should be coupled to real service usage, not financial speculation; and demand concentration should be monitored alongside S2R as a separate health metric.

**Subscription-based burns produce lower concentration than carrier contracts.** GEODNET's subscription model (Section 5.2.5) generates individually small but broadly distributed burns, contrasting with Helium's carrier-contract dependence. Where service pricing must be predictable for enterprise buyers, a dual-token or credit-denominated model reduces reflexivity risk.

**Spatial staking as frontier coverage mechanism.** GEODNET's SuperHex system (Section 5.2.5) ties staking requirements to geographic coverage gaps, creating an economic incentive for frontier deployment without protocol-managed hardware

subsidies. This represents a design primitive unavailable to non-geographic protocols.

Unit economics subsidy decomposition. The DePIN vs. Web2 comparison (Table 5) shows pricing 1.4 to 80 times below incumbents, but three of five domains carry medium or high subsidy risk. Service verification remains the hardest design problem: token incentives must reward actual service delivery, not merely asset deployment, as the Helium HIP-147 transition demonstrates (Section 5.1.1).

Subsidy taper creates fiscal cliffs. DePIN protocols face domain-specific taper risks as token emissions decline. Subsidy taper scenarios for each domain are developed in Appendix C.

Service incentives must be verifiable. The Helium HIP-147 transition from coverage-based to service-weighted rewards illustrates the core DePIN design challenge. The S2R trajectory (Table 3) shows this shift's fiscal impact: burn-to-issuance ratios rose from 0.01 to above 1.0 as rewards tracked real data transfer.

Forking asymmetry strengthens voice. DePIN's physical assets create forking asymmetry: software can be forked freely, but hotspots and storage nodes cannot easily switch networks. This makes contestation infrastructure (appeals, mediation, constitutional protections) more important than in pure software protocols.

## 6.2 The Subtraction Test [Proposal]

Deploying all recommended mechanisms simultaneously would reproduce the premature complexity anti-pattern identified in Section 6.4. The design implications above map institutional conditions to mechanisms, but the following filter operationalizes a principle of mechanism parsimony: before adding any governance or tokenomic mechanism, the designer should answer three questions.

(1) What behavior does this mechanism create? State the intended behavioral change in one sentence. If it cannot be stated simply, the mechanism may not be understood by participants, violating the transparency principle. Example: "Time-locked staking reduces reflexive selling during price drops."

(2) What misunderstanding will this mechanism create? Identify the most likely misinterpretation. Every mechanism creates a mental model in participants' minds; if the intended model and the received model diverge, the mechanism will produce unintended behavior. Example: participants may interpret staking locks as illiquidity risk instead of commitment signaling, triggering the opposite of the intended effect.



(3) What is the smallest version that demonstrates the behavior without requiring belief? Strip the mechanism to its testable core. If the mechanism requires participants to trust that future conditions will materialize (e.g., “burns will increase when adoption grows”), it is not yet testable and should be deferred until the prerequisite condition exists. The Helium trajectory illustrates this: HIP-147’s shift to service-based rewards worked because network usage already existed; it would have failed in 2020 when the network had coverage but no demand.

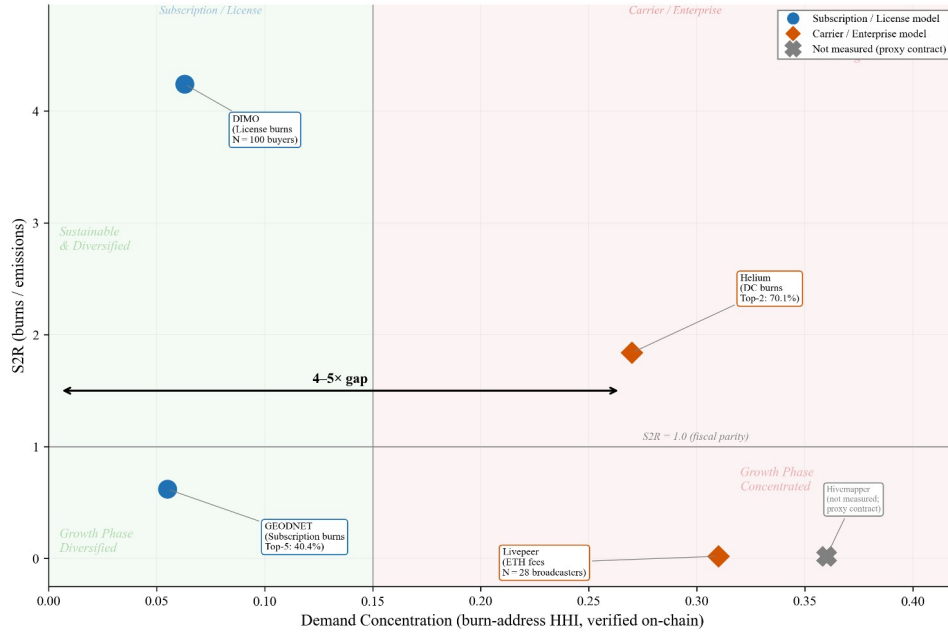
This filter is the practical complement to the design implications. Section 6.1 identifies what to deploy under which conditions; the subtraction test says when to stop. Together they encode the core design philosophy of this paper: institutional legitimacy requires not only the right mechanisms but the right sequence, earning complexity through demonstrated need, adding mechanisms only after the prior layer is tested. For GeoDePIN protocols specifically, this means spatial reward routing should be introduced only after the network has sufficient coverage data to know where deployment creates value, not before.

### 6.3 DePIN Health Dashboard [Inferred]

Drawing on the empirical data assembled in Section 5, Table 8 proposes a set of 12-month health targets for DePIN protocols entering the growth phase. These are not empirically derived thresholds but normative benchmarks grounded in observed ranges across operating protocols. They are intended as a practitioner tool: a dashboard that protocol teams and governance participants can use to assess whether incentive structures are on track.

Demand concentration as a second diagnostic axis. The Helium “Who Burns?” analysis (Section 5.1) reveals that S2R and demand distribution are independent dimensions of protocol health. S2R measures fiscal sustainability (are burns offsetting issuance?); demand concentration measures counterparty resilience (is demand broad or captive?). The interaction produces four regimes: (1) High S2R / Low demand concentration = ideal, where burns exceed issuance and demand is distributed; (2) High S2R / High demand concentration = “single-buyer fragility,” where the protocol is fiscally healthy but depends on a few large buyers whose departure would collapse the burn rate; (3) Low S2R / Low demand concentration = “diffuse but weak demand,” where many participants use the network but not enough to offset issuance; (4) Low S2R / High demand concentration = “fragile and captive,” the worst case.

Helium in H2 2025 sits in regime 2: S2R crossed parity (1.84) but the top-5 burn signers account for approximately 90% of total burns. Analysts should apply this framing to every GeoDePIN with a burn mechanism. We propose tracking Top-5 burn share alongside S2R in the health dashboard below.



Source: Dime Analytics (Q6942532, Q6942570, Q6917762, Q6942646). All values verified from on-chain burn/fee data, Jan-Mar 2026. Helium excluded: 96.1% of burns from single relay contract.

**Figure 4. Demand-concentration  $\times$  S2R regime matrix for four GeoDePIN protocols.**  
**Quadrant boundaries: S2R = 1.0 (horizontal), HHI = 0.10 (vertical).**

Three metrics anchor the dashboard (Table 6). S2R  $\geq 0.35$  means at least one-third of emissions are offset by usage-linked burns or locks. Helium far exceeds this target (S2R 1.84 as of February 2026 [33]), driven by carrier revenue burns and post-halving issuance reduction; the health dashboard should track whether S2R remains above 0.35 over rolling 6-month windows, not single months.

Token stickiness (second metric):  $\geq 60\%$  means the majority of reward recipients hold their tokens beyond the initial vesting period; this is aspirational given that early Helium hosts widely sold rewards during the 2021–2022 speculative period. Owner-user overlap  $\geq 20\%$  signals a meaningful core of participants who both provide and consume the service.

Demand concentration (third metric): If Top-5 burn share exceeds 80% for 6 consecutive months, activate counterparty risk mitigations (new enterprise buyers, retail usage bundles, corridor-level incentives). This trigger converts the “Who Burns?” caveat into a general governance control.

Across four DePIN protocols with verified on-chain demand data, subscription-based burn models (GEODNET HHI = 0.055, DIMO HHI = 0.063) produce approximately four to five times less demand concentration than carrier or enterprise-contract models (Helium HHI = 0.27, Livepeer HHI = 0.31), a pattern consistent regardless of absolute revenue level or buyer count.

The demand concentration diagnostic draws on a well-documented risk in corporate finance and supply chain management. Resource Dependence Theory [43] predicts that asymmetric reliance on a small number of buyers grants those buyers disproportionate influence over the platform’s strategic direction, a structural vulnerability that empirical studies confirm increases cost of equity capital and induces strategic short-termism [44]. Helium’s burn-signer HHI of 0.27, with two addresses controlling 70% of Data Credit burns, falls within the range this literature identifies as introducing material counterparty risk. GEODNET’s subscription model (HHI = 0.055) and DIMO’s license model (HHI = 0.063) produce four to five times less demand concentration, consistent with Resource Dependence Theory’s prediction that diversified revenue sources reduce buyer leverage. The Top-5 threshold of 80% proposed above translates this corporate finance insight into a protocol-specific monitoring trigger. The demand concentration diagnostic also captures an oligopsony risk identified in industrial organization theory: when a small number of buyers account for the majority of network revenue, their bargaining power compresses operator margins and constrains protocol governance, a dynamic that subscription models structurally mitigate through buyer fragmentation.

| Metric                   | Target       | Helium Actual (Feb 2026)    | Basis                                   |
|--------------------------|--------------|-----------------------------|-----------------------------------------|
| S2R (burns only)         | ≥ 0.35       | 1.84 (far exceeds)          | >1/3 of emissions offset by usage sinks |
| Token stickiness         | ≥ 60% at 4wk | Unknown (improving)         | Standard retention benchmark            |
| Owner-user overlap       | ≥ 20%        | Unknown (proxy only)        | Minimum alignment signal                |
| p95 node uptime          | ≥ 95%        | ~90% (dense), ~75% (sparse) | Infrastructure reliability floor        |
| Post-incentive retention | ≥ 60% at 4wk | Unknown                     | Anti-mercenary threshold                |

*Table 8. DePIN health dashboard metrics with proposed 12-month targets, Helium actuals (February 2026), and empirical basis.*

*Table 6. Health dashboard: proposed 12-month benchmarks.*

| Protocol   | Sink Type                  | S2R (burns) | S2R (burns+locks) | Period            | Source                                                                     | Demand HHI |
|------------|----------------------------|-------------|-------------------|-------------------|----------------------------------------------------------------------------|------------|
| Helium     | Full DC burn               | 0.02-1.84   | higher            | Apr 2023–Feb 2026 | Dune (on-chain)                                                            | 0.27       |
| Filecoin   | Base fee burn + collateral | ~0.045      | 0.14-0.41         | Jun 2023          | Deep Research                                                              | —          |
| Livepeer   | None (fees in ETH)         | ~0          | ~0                | Q1 2025           | DeFi Llama                                                                 | 0.31       |
| Hivemapper | Map credit burn            | 0.183       | 0.183             | Feb 2026          | Dune Analytics (Feb 2026); burn-and-mint active but not queryable via Dune | —          |
| GEODNET    | Subscription burn          | 0.62        | 0.62              | Feb 2026          | Dune Solana (on-chain)                                                     | 0.055      |
| DIMO       | License burn               | 4.24        | —                 | Feb 2026          | Dune Polygon (on-chain)                                                    | 0.063      |
| IoTeX      | Gas fee burn               | N/A         | N/A               | Feb 2026          | IoTeX L1 not indexed by Dune; gas fee burn active on native chain          | —          |

| Protocol  | Sink Type           | S2R (burns) | S2R (burns+locks) | Period   | Source                                                                   | Demand HHI |
|-----------|---------------------|-------------|-------------------|----------|--------------------------------------------------------------------------|------------|
| WeatherXM | Subscription burn   | N/A         | N/A               | Feb 2026 | Subscription burn active; volumes too small for reliable S2R computation | —          |
| Grass     | None (staking only) | ~0          | ~0                | Feb 2026 | Public docs                                                              | —          |
| Anyone    | Stake/lock          | ~0          | ~0                | Feb 2026 | Public docs                                                              | —          |

*Table 9. Cross-protocol S2R and demand concentration summary across the 13-protocol sample. Column 'S2R (burns)' uses the burns-only definition; 'S2R (burns+locks)' includes token lock-up mechanisms where applicable. Demand HHI computed where per-address burn data is available.*

Table 9 consolidates the S2R and demand concentration readings from Section 5 across the 13-protocol sample.

*Table 7. Cross-protocol S2R comparison (9 DePIN protocols).*

## 6.4 Risks and Trade-offs

Genuine trade-offs pervade every design choice above. No institutional design can simultaneously maximize all normative values; the art is in calibrating the balance for specific contexts.

**Efficiency vs. Inclusivity.** Adaptive efficiency (automated or expert-driven decisions) conflicts with deliberative inclusion (slower, participatory process). We saw this in Ethereum's EIP-1559 [5] fee mechanism, where automated base fee adjustment coexists with governance deliberation over parameter changes. The resolution is domain-specific: categorize decisions by urgency and reversibility, applying fast-track processes for routine parameters and full deliberation for constitutional changes.

**Decentralization vs. Coherence.** Polycentric governance [14] can tip into fragmentation. Helium's subnetwork architecture shows the mitigation: IoT and Mobile networks share a common governance framework but maintain separate reward pools, achieving local autonomy within coordinated boundaries. The constraint is that some domains demand uniform standards (security, interoperability), limiting how far polycentricity can extend.

**Transparency vs. Privacy/Security.** Transparency requirements clash with operational security and personal privacy. Full on-chain transparency of governance votes enables accountability but also enables vote-buying and coercion. The resolution is contextual: publish governance rules and aggregate outcomes publicly while protecting individual vote privacy where coercion risk is high.

**Innovation vs. Stability.** Experimentation is essential for institutional learning but risky for systems managing real value. The health dashboard (Section 6.3) addresses this by defining trigger conditions: innovate freely above safety thresholds, but invoke conservative governance when metrics breach critical levels.

**Token Incentives vs. Intrinsic Motivation.** Tokenomics can crowd out intrinsic motivation, as the Uniswap airdrop case illustrates (90.8% claim rate but only 7% long-term retention [8], [24]). The design implication is to use tokens for coordination and accountability, with reputation and recognition as complementary motivational levers, and to invest in reputation systems and community recognition alongside financial incentives.

**Scale Limitations.** Many mechanisms work in small communities; as protocols scale, direct participation becomes impractical and delegation concentration increases. The mitigation is layered governance: local autonomy for routine decisions, representative governance for system-wide issues, and constitutional protections at all levels.

**Premature Mechanism Complexity.** S2R data reveals an anti-pattern: protocols that deploy elaborate tokenomics before establishing demand-coupled burns show S2R near zero (Grass, UpRock, early Helium pre-2024), while protocols that earned complexity through iteration achieve meaningful S2R (Helium 1.84, DIMO 4.24, GEODNET 0.62). A mechanism too complex for ordinary participants to verify violates the transparency principle regardless of whether the code is open-source. The implication is that mechanism designers should treat complexity as a cost, not a feature, and deploy new mechanisms only after the prior layer has been tested and understood by participants.

## 7. CONCLUSION

### 7.1 Summary

The S2R data across 13 protocols reveal a three-tier sustainability hierarchy for DePIN: Helium (1.84) and DIMO (4.24) have crossed fiscal parity through demand-coupled burns; GEODNET (0.62) and Hivemapper (0.183) occupy a growth-phase middle tier with confirmed burn activity but continued subsidy dependence; and the remaining protocols cluster near S2R zero, the sector default. The Helium longitudinal case (Table 3) demonstrates the trajectory: demand-coupled burns can drive S2R above fiscal parity when paired with service-based reward transitions (HIP-147), but demand concentration (top-5 signers at approximately 90% of burns) creates a shadow risk invisible to aggregate metrics. Governance

concentration in GeoDePIN tokens (HHI 0.034–0.593) materially exceeds DeFi benchmarks (0.028–0.174), consistent with the 13-protocol cross-section. The four-loop design architecture (economic, operational, integrity, governance), grounded in institutional design principles, and the GeoDePIN vs. DeVIN taxonomy (Table 2) organize these findings into testable structure.

## 7.2 Contributions

GeoDePIN protocols face institutional design pressures that generic DePIN analysis obscures: location-dependent verification, hardware lock-in that raises exit costs, and subsidy cliffs that force fiscal sustainability. Four artifacts address these pressures: (1) a GeoDePIN vs. DeVIN taxonomy with testable predictions about governance concentration, verification complexity, and subsidy sensitivity; (2) a demand-concentration diagnostic (the "Who Burns?" analysis and the S2R x demand-concentration 2x2 framework); (3) a design playbook keyed to observable protocol conditions with a subtraction test for minimum viable mechanism complexity; and (4) a health dashboard with demand-concentration guardrails (Top-5 > 80% AND Top-1 > 35% for 6 months triggers diversification action).

## 7.3 Limitations

The governance concentration cross-section is a February 2026 snapshot; concentration trajectories over time would strengthen causal claims. Unit economics use posted prices ( $N = 1$  per service domain); we estimate subsidy decomposition. The Helium S2R trajectory is observational: HIP-147 was not randomized, and confounders cannot be fully separated. Several protocols remain data-insufficient; their metrics are unmeasured, not absent. The sample includes only protocols that have survived to 2026; failed or abandoned projects may exhibit different patterns. Design recommendations generalize from Helium (the primary longitudinal case) to other GeoDePIN protocols by analogy; direct replication across multiple protocols would strengthen external validity.

The S2R metric as currently specified measures burns against mining emissions; it does not capture vesting-driven dilution from team, investor, and ecosystem unlocks, which for GEODNET represented 54% of total supply by Q2 2025. A comprehensive fiscal sustainability metric would incorporate all supply sources; this extension is specified as a future research direction.

S2R measures fiscal sustainability and demand coupling but does not capture participant-level welfare: an S2R above 1.0 can coexist with operator financial distress when token-to-fiat conversion factors are adverse [48]. The San Jose pilot [50] demonstrates this decoupling during the pre-parity regime. Participant-level return-on-investment analysis (matching on-chain reward data with off-chain

hardware costs, electricity consumption, and maintenance labor) remains a measurement priority.

More broadly, DePIN protocols claim to deliver public infrastructure goods (cheaper wireless, broader positioning coverage, fresher mapping data), yet every evaluation metric in the present study and in the existing literature measures token economy health rather than whether participants' quality of life actually improved. This welfare measurement gap is a structural property of on-chain data architectures, not merely a data limitation of the present study: protocols instrument token flows but not service outcomes. Connecting fiscal sustainability to participant welfare requires instrumentation that does not yet exist in the DePIN ecosystem.

#### 7.4 Future Work

Three additional research directions emerge from the empirical findings. HIP-147 constitutes a named, dated governance action suitable for formal event study analysis, with strong pre/post data for S2R and weaker data for service quality outcomes. Simulation evidence [29] suggests a Slashing-to-Burn Ratio as a third diagnostic axis; empirical validation across operating protocols is needed.

Agent-based simulation of DePIN market dynamics using large language models (Liu and Schweitzer, 2025) offers a complementary methodology for stress-testing the health dashboard thresholds proposed here, particularly for modeling how concentrated buyers respond to emission halvings.

Longitudinal analysis of token economic failure patterns identifies tokenomics-market fit disconnect as a primary failure mode (Belkin, 2026), validating the health dashboard's emphasis on trajectory monitoring over point-in-time assessment.

Beyond these, two extensions have highest priority. First, longitudinal governance tracking: monthly HHI/Gini snapshots would convert the cross-section into a panel and enable deconcentration analysis. Second, capability measurement: most protocols do not instrument real human outcomes; partnering with protocols to collect participant-level data would fill the most important gap. The burn-concentration finding (top-5 signers at 40.4% versus Helium's 90%) opens a new research question: whether subscription-based demand architectures systematically produce more distributed burn patterns than carrier-contract models, testable as additional GeoDePIN protocols mature.

The combination of verified demand concentration data across four protocols with Resource Dependence Theory's causal predictions generates a testable hypothesis: DePIN protocols with subscription-based burn models (demand HHI below 0.10) should exhibit more stable S2R trajectories across market cycles than carrier or



enterprise-contract models (demand HHI above 0.20), because lower demand concentration reduces the probability of simultaneous buyer exit. Longitudinal tracking of both S2R and demand HHI across halving events would provide the first empirical test of whether burn-model architecture determines fiscal resilience. Shapley value analysis of demand concentration would quantify whether the top burn addresses' probability of being pivotal in fiscal sustainability exceeds their nominal burn share. Helium's S2R = 1.0 threshold defines a binary characteristic function: coalitions of burn addresses whose combined contribution meets emissions earn value 1, others earn 0. The top-2 addresses (70.1% of burns) are not merely large contributors; they are the margin between self-sustaining and subsidy-dependent regimes, a distinction HHI captures in magnitude but not in pivotality.

## **Appendix A: Practitioner Brief**

The Practitioner Brief translates the design architecture and playbook into an actionable checklist for token designers and protocol founders, including a token justification litmus test, the four-loop institutional design system, DePIN-specific reward decomposition guidance, reflexivity-aware fiscal policy rules, and a KPI dashboard specification. Available as Supplementary Material S1.

## **Appendix B: Policy Brief**

The Policy Brief provides a governance and consumer-protection framework for regulators engaging with tokenized institutions, including a proposed Tokenized Institution Disclosure Template, oversight metrics (economic stability, governance legitimacy, service quality, privacy), safe-harbor criteria tied to governance quality, and a pragmatic implementation pathway. Available as Supplementary Material S2.

## **Appendix C: Subsidy Taper Scenarios**

A central question for DePIN sustainability is what happens to service pricing when token emissions decline. The unit economics in Section 5.4 show DePIN pricing 1.4–80x below Web2 incumbents, but three of five domains carry Medium or High subsidy risk. The following qualitative scenarios assess each domain's resilience to a 50% emission reduction:

*Filecoin (Low-Medium risk):* Miners already bid competitively for storage deals and earn fees directly from clients. A 50% emission reduction would increase the FIL cost of storage per GB-month, but the collateral requirement already filters for economically viable operators. Storage pricing would likely rise but remain below AWS S3 given Filecoin's lower infrastructure overhead.



Livepeer (Medium risk): Orchestrators earn ETH fees directly from clients (\$3,071/day average, February 2026). LPT staking rewards supplement income but are not the primary revenue source. A 50% LPT emission reduction would reduce staking yields, causing some orchestrators to exit, but the direct fee revenue model is structurally sound.

*Helium Mobile (High risk):* Hotspot operators are primarily compensated through HNT rewards, not subscriber fees. The \$30/month unlimited plan is priced well below T-Mobile's \$70, but the gap is funded by token emissions. A 50% emission reduction would require either raising the plan price, increasing subscriber density per hotspot, or relying on operators who are also subscribers (owner-user overlap). This is the domain where sustainability is most uncertain and where the S2R metric is most critical as an early warning indicator.

*Hivemapper (Medium risk):* On-chain burn data confirm active demand (S2R 0.183, February 2026), but burns offset less than one-fifth of issuance, leaving the mapping network in its growth phase. Sustainability depends on achieving sufficient commercial demand for fresh street-level imagery to drive S2R toward parity as token rewards taper.

These scenarios do not constitute predictions but rather inform the monitoring cadence recommended above: protocols with High subsidy risk should track S2R and stickiness monthly, while those with Low risk can review quarterly.

## Biography

Zach Zukowski is a researcher studying the intersection of institutional design, political philosophy, and decentralized systems. His work applies normative frameworks from political economy and constitutional theory to the design and governance of tokenized infrastructure networks. His research focuses on how token incentive systems function as institutional architectures encoding rights, duties, and enforcement mechanisms analogous to public policy instruments.

## Declaration of Competing Interest

The author served as Senior Investment Analyst at Borderless Capital through February 2026 and retains no decision-making role, carried interest, or position-dependent compensation with the firm. The author holds personal token positions in three protocols analyzed in this paper: HNT (Helium), DIMO, and GEOD (GEODNET); positions are small and were established prior to the research period. The research, analysis, and conclusions presented are the author's own; no sponsor,

portfolio holding, or third party had editorial input or review rights prior to submission.

## Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

## CRediT Authorship Contribution Statement

Zach Zukowski: Conceptualization, Methodology, Data curation, Formal analysis, Writing – original draft, Writing – review & editing, Visualization.

## Data Availability

The data supporting the findings of this study are publicly available from on-chain sources (Dune Analytics, Helius DAS API, Blockscout) via the queries specified in Section 3.1. Processed datasets and replication scripts are archived at the public repository [https://github.com/Research-Publications-and-Data/Tokenomics-As-Institutional\\_Design](https://github.com/Research-Publications-and-Data/Tokenomics-As-Institutional_Design) (tag v1.0.0-frontiers-submission for the companion paper [34]; B3 replication materials will be tagged at acceptance).

## ORCID

Zach Zukowski: <https://orcid.org/0009-0006-3642-2450>

## References

- [1] S. Aramonte, W. Huang, and A. Schrimpf, "DeFi risks and the decentralisation illusion," *BIS Quart. Rev.*, pp. 21-36, Dec. 2021.
- [2] D. Bane and S. Gala, "State of DePIN 2024," Messari, Jan. 2025. [Online]. Available: <https://messari.io/report/state-of-depin-2024>
- [3] T. Barbereau, R. Smethurst, O. Papageorgiou, J. Sedlmeir, and G. Fridgen, "Decentralised finance's timocratic governance: The distribution and exercise of tokenised voting rights," *Technol. Soc.*, vol. 73, p. 102251, 2023.
- [4] J. M. Buchanan and G. Tullock, *The Calculus of Consent: Logical Foundations of Constitutional Democracy*. Ann Arbor, MI: Univ. of Michigan Press, 1962.
- [5] V. Buterin, E. Conner, R. Dudley, M. Slipper, and I. Norder, "EIP-1559: Fee market change for ETH 1.0 chain," *Ethereum Improvement Proposals*, 2019.

- [6] CoinGecko, "Cryptocurrency market data," 2026. [Online]. Available:  
<https://www.coingecko.com>
- [7] DeFi Llama, "DeFi protocol analytics," 2026. [Online]. Available:  
<https://defillama.com>
- [8] Dune Analytics, "On-chain governance and burn data," 2026. [Online]. Available:  
<https://dune.com>
- [9] Helium, "HIP-147: Mobile data eats first," Helium Improvement Proposal, Aug. 20, 2025. [Online]. Available:  
<https://github.com/helium/HIP/blob/main/0147-mobile-data-eats-first.md>
- [10] L. Hurwicz, "The design of mechanisms for resource allocation," *Amer. Econ. Rev.*, vol. 63, no. 2, pp. 1-30, 1973.
- [11] R. B. Myerson, "Optimal auction design," *Math. Oper. Res.*, vol. 6, no. 1, pp. 58-73, 1981.
- [12] M. Nadler and F. Schar, "Decentralized finance, centralized ownership? An iterative mapping process to measure protocol token distribution," *arXiv:2012.09306*, 2020.
- [13] D. C. North, *Institutions, Institutional Change and Economic Performance*. Cambridge, U.K.: Cambridge Univ. Press, 1990.
- [14] E. Ostrom, *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge, U.K.: Cambridge Univ. Press, 1990.
- [15] Protocol Labs, "Filecoin: A decentralized storage network," Whitepaper, Jul. 2017. [Online]. Available: <https://filecoin.io/filecoin.pdf>
- [16] J.-C. Rochet and J. Tirole, "Platform competition in two-sided markets," *J. Eur. Econ. Assoc.*, vol. 1, no. 4, pp. 990-1029, 2003.
- [17] T. Roughgarden, "Transaction fee mechanism design," *ACM SIGecom Exchanges*, vol. 19, no. 1, pp. 52-55, 2021.
- [18] F. Schar, "Decentralized finance: On blockchain- and smart contract-based financial markets," *Federal Reserve Bank of St. Louis Rev.*, vol. 103, no. 2, pp. 153-174, 2021.
- [19] S. Voshmgir, *Token Economy: How the Web3 Reinvents the Internet*, 2nd ed. Vienna, Austria: Token Kitchen, 2020.

- [20] S. M. Werner, D. Perez, L. Gudgeon, A. Klages-Mundt, D. Harz, and W. J. Knottenbelt, "SoK: Decentralized finance (DeFi)," in Proc. 4th ACM Conf. Advances Financial Technol., 2022, pp. 30-46.
- [21] R. K. Yin, Case Study Research and Applications: Design and Methods, 6th ed. Thousand Oaks, CA: Sage, 2018.
- [22] C. Catalini and J. S. Gans, "Some simple economics of the blockchain," Commun. ACM, vol. 63, no. 7, pp. 80-90, Jul. 2020.
- [23] S. T. Howell, M. Niessner, and D. Yermack, "Initial coin offerings: Financing growth with cryptocurrency token sales," Rev. Financial Stud., vol. 33, no. 9, pp. 3925-3974, Sep. 2020.
- [24] R. Feichtinger, R. Fritsch, Y. Vonlanthen, and R. Wattenhofer, "The hidden shortcomings of (D)AOs: An empirical study of on-chain governance," in Proc. Int. Conf. Financial Cryptography Data Security, 2023, pp. 165-185.
- [25] O. Rikken, M. Janssen, and Z. Kwee, "Governance impacts of blockchain-based decentralized autonomous organizations: An empirical analysis," Policy Design Practice, vol. 6, no. 4, pp. 465-487, 2023.
- [26] O. E. Williamson, The Economic Institutions of Capitalism: Firms, Markets, Relational Contracting. New York, NY: Free Press, 1985.
- [27] L. W. Cong, D. Rabetti, C. C. Y. Wang, and Y. Yan, "Centralized governance in decentralized organizations," Working Paper (Draft: August 2025), 2025.
- [28] D. Bane and S. Gala, "State of DePIN 2025," Messari and Escape Velocity, Jan. 2026. [Online]. Available: <https://messari.io/report/state-of-depin-2025>
- [29] Z. Zukowski, "Tokenomics as systems engineering: Protocols, labor, and the design of autonomous economies," SSRN Working Paper 6364158, 2026. doi: 10.2139/ssrn.6364158.
- [30] N. Arnosti and S. M. Weinberg, "Bitcoin: A natural oligopoly," Management Science, vol. 68, no. 7, pp. 4755-4771, 2022.
- [31] L. van Vulpen and S. Jansen, "Decentralized autonomous organization design for the commons and the common good," Frontiers in Blockchain, vol. 6, 1257313, 2023.

- [32] Blockworks, "GEODNET analytics dashboard," 2026. [Online]. Available: <https://blockworks.com/analytics/geodnet> (Retrieved Apr. 2026).
- [33] Blockworks, "Helium financials dashboard," 2026. [Online]. Available: <https://blockworks.com/analytics/helium/helium-financials> (Retrieved Apr. 2026).
- [34] Z. Zukowski, "Governance concentration beyond token allocation: An institutional design analysis of DePIN and DeFi," SSRN Working Paper 6599278, 2026. doi: 10.2139/ssrn.6599278.
- [35] M. C. Ballandies, H. Wang, A. C. C. Law, J. C. Yang, C. Gosken, and M. Andrew, "A taxonomy for blockchain-based decentralized physical infrastructure networks (DePIN)," in 2023 IEEE 9th World Forum on Internet of Things (WF-IoT), 2023.
- [36] M. S. Andrew and M. C. Ballandies, "Are you a DePIN? A decision tree to classify decentralized physical infrastructure networks," in Crypto Valley Conference (CVC), 2025.
- [37] O. Akcin, K. Poolla, and P. Varaiya, "A control theoretic approach to infrastructure-centric blockchain tokenomics," arXiv:2210.12881, 2022.
- [38] U. Kalabic, T. Keviczky, and R. Poonacha, "Burn-and-mint tokenomics: Deflation and strategic incentives," in 1st Int. Workshop on DePIN, IEEE WF-IoT, 2023.
- [39] J. Soria Ruiz-Ogarrio, J. Moya Velasco, and C. Estevez-Mendoza, "The road to decentralization: Optimal coverage on decentralized physical infrastructure networks (DePIN)," Finance Research Letters, 2026.
- [40] M. Corn, S. Mahon, P. Bellotti, and J. Foretia, "Decentralized physical infrastructure networks (DePINs) for solar energy: The impact of network density on forecasting," Forecasting (MDPI), 2025.
- [41] M. M. Alshater, "Decentralized physical infrastructure networks (DePIN) tokenomics," Frontiers in Blockchain, 2026.
- [42] D. Jagtap, F. Garg, N. Singh, W. Kaiser, and M. Srivastava, "Federated infrastructure: Usage, patterns, and insights from "The People's Network,"" in Proc. ACM Internet Measurement Conf. (IMC), 2021.
- [43] J. Pfeffer and G. R. Salancik, *The External Control of Organizations: A Resource Dependence Perspective*. New York, NY: Harper & Row, 1978.
- [44] D. Dhaliwal, J. S. Judd, M. Serfling, and S. Shaikh, "Customer concentration risk and the cost of equity capital," J. Account. Econ., vol. 61, no. 1, pp. 23–48, 2016.

- [45] Messari, "GEODNET Quarterly Report Q2 2025," Messari Research, Jul. 2025. [Online]. Available: <https://messari.io/research> (Retrieved Apr. 2026).
- [46] L. S. Shapley, "A value for n-person games," in H. W. Kuhn and A. W. Tucker (Eds.), *Contributions to the theory of games, Volume II* (Annals of Mathematics Studies 28, pp. 307–317). Princeton, NJ: Princeton University Press, 1953.
- [47] I. Ferreras, *Firms as Political Entities: Saving Democracy through Economic Bicameralism*. Cambridge, U.K.: Cambridge Univ. Press, 2017.
- [48] A. Sen, *Development as Freedom*. New York, NY: Alfred A. Knopf, 1999.
- [49] S. Motepalli, N. Garg, G. Zhang, and H.-A. Jacobsen, "GPoS: Geospatially-aware proof of stake," arXiv:2511.02034, 2025.
- [50] StateScoop, "San Jose, Calif., partnership with crypto company Helium was a flop," Sep. 26, 2022. [Online]. Available: <https://statescoop.com/san-jose-calif-partnership-crypto-company-helium-flopped/>
- [51] A. Lijphart, *Patterns of Democracy: Government Forms and Performance in Thirty-Six Countries*, 2nd ed. New Haven, CT: Yale Univ. Press, 2012.
- [52] S. Motepalli, N. Garg, G. Zhang, and H.-A. Jacobsen, "Decentralization in PoS blockchain consensus: Quantification and advancement," arXiv:2504.14351, 2025.
- [53] Z. Lin, T. Wang, L. Shi, S. Zhang, and B. Cao, "Decentralized physical infrastructure networks (DePIN): Challenges and opportunities," *IEEE Network*, 2024. doi: 10.1109/MNET.2024.3487924.
- [54] B. Petrenj and P. Trucco, "Decentralized physical infrastructure networks: A catalyst for critical infrastructure resilience?" in *Proc. 33rd European Safety and Reliability & 33rd Society for Risk Analysis Europe Conf. (ESREL SRA-E 2025)*, Research Publishing, Singapore, 2025. doi: 10.3850/978-981-94-3281-3\_ESREL-SRA-E2025-P3352-cd.
- [55] Y. Wu and F. Zhu, "A paradox of utility tokens," *Management Science*, vol. 68, no. 6, 2022. doi: 10.1287/mnsc.2022.4314.
- [56] J. Millionis, J. Ernstberger, J. Bonneau, S. D. Kominers, and T. Roughgarden, "Incentive-compatible recovery from manipulated signals, with applications to decentralized physical infrastructure," arXiv:2503.07558, 2025.
- [57] J. von der Assen, C. Killer, A. De Carli, and B. Stiller, "Performance analysis of decentralized physical infrastructure networks and centralized clouds," in *Proc.*

IEEE Int. Conf. Blockchain and Cryptocurrency (ICBC), 2024. doi: 10.1109/ICBC59979.2024.10634394.

[58] P. Rad, M. Jozani, G. Zanella, M. Safaei Pour, and K. Abhari, "Reimagining the sharing economy through blockchain: The case of Helium's decentralized wireless network," in Proc. 58th Hawaii Int. Conf. System Sciences (HICSS), 2025. doi: 10.24251/HICSS.2025.674.

[59] H. Fiegenbaum, B. Azegele, and S. Seider, "Decentralized internet-of-things networks for environmental data pipelines: Opportunities and challenges in a community-based urban green space restoration project in Nairobi, Kenya," Frontiers in Blockchain, 2026. doi: 10.3389/fbloc.2025.1626695.

[60] A. Shapira, "Helium, touted as crypto telecom of the future, brings in just \$6,500 a month in usage fees," Mashable, Jul. 29, 2022. [Online]. Available: <https://mashable.com/article/helium-crypto-barely-anyone-uses>

[61] A. Haleem, "Helium 5G network growth update," Helium Foundation Blog, 2022. [Online]. Available: <https://blog.helium.com>